

Adaptive Incremental Nonlinear Dynamic Inversion for Attitude Control of Micro Air Vehicles

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DOI: 10.2514/1.G001490

Incremental nonlinear dynamic inversion is a sensor-based control approach that promises to provide high-performance nonlinear control without requiring a detailed model of the controlled vehicle. In the context of attitude control of micro air vehicles, incremental nonlinear dynamic inversion only uses a control effectiveness model and uses estimates of the angular accelerations to replace the rest of the model. This paper provides solutions for two major challenges of incremental nonlinear dynamic inversion control: how to deal with measurement and actuator delays, and how to deal with a changing control effectiveness. The main contributions of this article are 1) a proposed method to correctly take into account the delays occurring when deriving angular accelerations from angular rate measurements; 2) the introduction of adaptive incremental nonlinear dynamic inversion, which can estimate the control effectiveness online, eliminating the need for manual parameter estimation or tuning; and 3) the incorporation of the momentum of the propellers in the controller. This controller is suitable for vehicles that experience a different control effectiveness across their flight envelope. Furthermore, this approach requires only very coarse knowledge of model parameters in advance. Real-world experiments show the high performance, disturbance rejection, and adaptiveness properties.

Nomenclature

b	=	width of the vehicle, m
I	=	identity matrix
I_r	=	moment of inertia matrix of the rotor, $\text{kg} \cdot \text{m}^2$
I_v	=	moment of inertia matrix of the vehicle, $\text{kg} \cdot \text{m}^2$
i	=	rotor index
k_1	=	force constant of the rotors, $\text{kg} \cdot \text{m}/\text{rad}$
k_2	=	moment constant of the rotors, $\text{kg} \cdot \text{m}^2/\text{rad}$
l	=	length of the vehicle, m
M_a	=	aerodynamic moment vector acting on the vehicle, $\text{N} \cdot \text{m}$
M_c	=	control moment vector acting on the vehicle, $\text{N} \cdot \text{m}$
M_r	=	moment vector acting on the propeller, $\text{N} \cdot \text{m}$
T_s	=	sample time of the controller, s
u	=	actuator input vector, rad/s
v	=	vehicle velocity vector, m/s
μ	=	adaptation rate diagonal matrix
Ω	=	vehicle angular rate vector, rad/s
$\dot{\Omega}$	=	angular acceleration vector, rad/s^2
ω	=	angular rate vector of the four rotors around the body z axis, rad/s
ω_i	=	angular rate vector of rotor i around each of the body axes, rad/s

I. Introduction

MICRO air vehicles (MAVs) have increased in popularity as low-cost lightweight processors and inertial measurement units have become available through the smartphone revolution. The inertial sensors allow stabilization of unstable platforms by feedback algorithms. Typically, the stabilization algorithm used for MAVs is simple proportional integral derivative (PID) control [1,2]. Problems

with PID control occur when the vehicle is highly nonlinear or when the vehicle is subject to large disturbances like wind gusts.

Alternatively, we could opt for a model-based attitude controller. A model-based controller that can deal with nonlinear systems is nonlinear dynamic inversion (NDI), which involves modeling all of the MAV's forces and dynamics. Theoretically, this method can remove all nonlinearities from the system and create a linearizing control law. However, NDI is very sensitive to model inaccuracies [3]. Obtaining an accurate model is often expensive or impossible with the constraints of the sensors that are carried onboard a small MAV.

The incremental form of nonlinear dynamic inversion (INDI) is less model-dependent and more robust. It has been described in the literature since the late 1990s [4,5], sometimes referred to as simplified [6] or enhanced [7] NDI. Compared to NDI, instead of modeling the angular acceleration based on the state and inverting the actuator model to get the control input, the angular acceleration is measured, and an increment of the control input is calculated based on a desired increment in angular acceleration. This way, any unmodeled dynamics, including wind gust disturbances, are measured and compensated. Because INDI makes use of a sensor measurement to replace a large part of the model, it is considered a sensor-based approach.

INDI faces two major challenges. First, the measurement of angular acceleration is often noisy and requires filtering. This filtering introduces a delay in the measurement, which should be compensated for. Second, the method relies on inverting and therefore modeling the controls. To achieve a more flexible controller, the control effectiveness should be determined adaptively.

Delay in the angular acceleration measurement has been a prime topic in INDI research. A proposed method to deal with these measurement delays is predictive filtering [8]. However, the prediction of angular acceleration requires additional modeling. Moreover, disturbances cannot be predicted. Initially, a setup with multiple accelerometers was proposed by Bacon and Ostroff [5] to measure the angular acceleration. This setup has some drawbacks because it is complex and the accelerometers are sensitive to structural vibrations. Later, they discussed the derivation of angular acceleration from gyroscope measurements by using a second-order filter [9]. To compensate for the delay introduced by the filter, Bacon and Ostroff [5] use a lag filter on the applied input to the system. We show in this paper that perfect synchronization of input and measured output can be achieved by applying the filter used for the gyroscope differentiation on the incremented input as well.

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用于微型飞行器姿态控制的自适应增量非线性动态反演

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DOI: 10.2514/1.G001490

增量非线性动态反演是一种基于传感器的控制方法, 它有望在无需详细受控飞行器模型的情况下提供高性能的非线性控制。在微型航空器姿态控制的背景下, 增量非线性动态反演仅使用控制有效性模型, 并利用角加速度的估计来替代其余的模型。本文为增量非线性动态反演控制的两个主要挑战提供了解决方案: 如何处理测量和执行器的延迟, 以及如何应对控制有效性的变化。本文的主要贡献是: 1) 提出了一种方法, 在从角速度测量中推导角加速度时正确考虑所发生的延迟; 2) 引入了自适应增量非线性动态反演, 它可以在线估计控制有效性, 从而无需手动参数估计或调节。

命名法

b = 车辆宽度, 米 I = 单位矩阵 I_r = 转子转动惯量矩阵, 千克·米² I_v = 车辆转动惯量矩阵, 千克·米² i = 转子索引 k_1 = 转子力常数, 千克·米/弧度 k_2 = 转子力矩常数, 千克·米²/弧度 l = 车辆长度, 米 M_a = 作用于车辆的气动力矩向量, 牛·米 M_c = 作用于车辆的控制力矩向量, 牛·米 M_r = 作用于螺旋桨的力矩向量, 牛·米 T_s = 控制器采样时间, 秒 u = 执行器输入向量, 弧度/秒 v = 车辆速度向量, 米/秒 μ = 适应率对角矩阵 Ω = 车辆角速度向量, 弧度/秒 $\underline{\Omega}$ = 角加速度向量, 弧度/秒² ω = 四个转子绕机体系 z 轴的角速度向量, 弧度/秒 ω_i = 转子 i 绕每个机体系轴的角速度向量, 弧度/秒

当车辆高度非线性或车辆受到大扰动如阵风时, 会发生 PID 控制问题。

或者, 我们可以选择基于模型的姿态控制器。能够处理非线性系统的基于模型的控制器的非线性动态反演 (NDI), 它涉及对微型无人机的所有力和动力学进行建模。理论上, 这种方法可以消除系统中的所有非线性, 并创建一个线性化的控制律。然而, NDI 对模型的不准确性非常敏感 [3]。在小型微型无人机所携带的传感器限制下, 获取准确的模型通常成本高昂或不可能。

非线性动态逆 (NDI) 的增量形式 (INDI) 依赖于模型较少, 更加稳健。自20世纪90年代末以来, 文献中已有相关描述[4,5], 有时被称为简化NDI[6]或增强NDI[7]。与NDI相比, INDI不是基于状态建模角加速度并反转执行器模型以获得控制输入, 而是测量角加速度, 并根据角加速度的期望增量计算控制输入的增量。通过这种方式, 任何未建模的动力学, 包括风阵扰动, 都可以被测量并补偿。由于INDI利用传感器测量来替代模型的大部分内容, 因此被认为是一种基于传感器的方法。

INDI 面临两个主要挑战。首先, 角加速度的测量通常存在噪声, 需要进行滤波。该滤波会引入测量延迟, 应予以补偿。其次, 该方法依赖于对控制进行反演, 因此需要对控制进行建模。为了实现更灵活的控制, 控制效果应当自适应地确定。

角加速度测量的延迟一直是INDI研究中的一个主要课题。应对这些测量延迟的一个提议方法是预测滤波 [8]。然而, 角加速度的预测需要额外的建模。此外, 干扰是无法预测的。最初, Bacon 和 Ostroff [5] 提出了一个由多个加速度计组成的装置来测量角加速度。该装置有一些缺点, 因为它结构复杂且加速度计对结构振动敏感。随后, 他们讨论了通过使用二阶滤波器 [9] 从陀螺仪测量中推导角加速度的方法。为了补偿滤波器引入的延迟, Bacon 和 Ostroff [5] 在系统的输入端使用滞后滤波器。本文中我们展示, 通过在增量输入上应用于陀螺仪微分的滤波器, 可以实现输入和测量输出的完美同步。

一、引言

微型空中车辆 (MAVs) 随着低成本轻量级处理器和惯性测量单元的出现而越来越受欢迎, 这些技术通过智能手机革命得以实现。惯性传感器通过反馈算法使不稳定的平台得以稳定。通常, 用于MAVs的稳定化算法是简单的比例-积分-微分 (PID) 控制 [1,2]。问题

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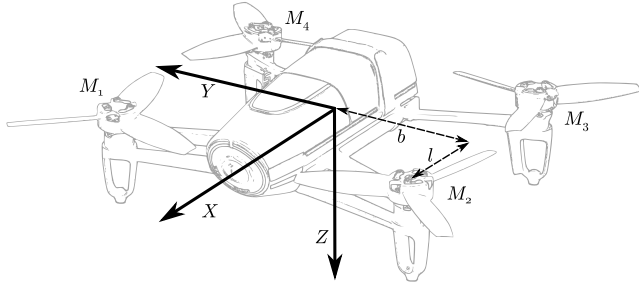


Fig. 1 Bebop quadcopter used in the experiments with axis definitions.

Other research focused on compensating delays in the inputs by using a Lyapunov-based controller design [10]. In this paper, we show that delayed inputs (actuator dynamics) are naturally handled by the INDI controller.

The control effectiveness is the sole model still required by INDI. The parameters can be obtained by careful modeling of the actuators and the moment of inertia or by analyzing the input output data from flight logs. However, even if such a tedious process is followed, the control effectiveness can change during flight. For instance, this can occur due to changes in flight conditions [11] or actuator damage [12]. To cope with this, we propose a method to adaptively determine the control effectiveness matrices.

In this paper, we present three main contributions: 1) a mathematically sound way of dealing with the delays originating from filtering of the gyroscope measurements, 2) the introduction of an adaptive INDI scheme, which can estimate the control effectiveness online, and 3) incorporation of propeller momentum in the controller design. These contributions are implemented and demonstrated on a Parrot Bebop quadrotor running the Paparazzi open-source autopilot software. This is a commercially available quadrotor, and the code is publicly available on Github.[§]

The presented theory and results generalize to other vehicles in a straightforward manner. We have applied this control approach successfully to a variety of quadrotors. Some of these MAVs were able to measure the rotational rate of the rotors (actuator feedback), but some did not have this ability. The INDI controller is believed to scale well to different types of MAVs like helicopter, multirotor, fixed wing, or hybrid.

The outline of this paper is as follows. First, a model of the MAV will be discussed in Sec. II. Second, Sec. III will deal with INDI and the analysis for this controller for a quadrotor. Section IV is about the adaptive extension of INDI. Finally, in Sec. V, the experimental setup is explained, followed by the results of the experiments in Sec. VI.

II. Micro Air Vehicle Model

The Bebop quadrotor is shown in Fig. 1 along with axis definitions. The actuators drive the four rotors, whose angular velocity in the body frame is given by $\omega_i = [\omega_{i_x}, \omega_{i_y}, \omega_{i_z}]$, where i denotes the rotor number. The center of gravity is located in the origin of the axis system, and the distance to each of the rotors along the X axis is given by l and along the Y axis by b .

If the angular velocity vector of the vehicle is denoted by $\Omega = [p, q, r]^T$ and its derivative by $\dot{\Omega}$, the rotational dynamics are given by Euler's equation of motion [13], more specifically the one that describes rotation. If we consider the body axis system as our coordinate system, we get Eq. (1) for the angular velocity of the vehicle:

$$I_v \dot{\Omega} + \Omega \times I_v \Omega = M \quad (1)$$

where M is the moment vector acting on the vehicle. If we consider the rotating propellers, still in the body coordinate system, we obtain

$$I_r \dot{\omega}_i + \Omega \times I_r \omega_i = M_{r_i} \quad (2)$$

where ω_i is the angular rate vector of the i th propeller in the vehicle body axes, and Ω is the angular rotation of the coordinate system, equal to the vehicle body rates. The rotors are assumed to be flat in the z axis, such that the inertia matrix I_r has elements that are zero: $I_{r_{xz}} = I_{r_{yz}} = 0$. Because the coordinate system is fixed to the vehicle, $I_{r_{xx}}, I_{r_{yy}},$ and $I_{r_{zz}}$ are not constant in time. However, as is shown later on, the terms containing these moments of inertia will disappear. Expanding Eq. (2) into its three components gives

$$\begin{aligned} I_{r_{xx}} \dot{\omega}_{i_x} - I_{r_{yy}} \Omega_z \omega_{i_y} - I_{r_{xy}} \Omega_z \omega_{i_x} + I_{r_{zz}} \Omega_y \omega_{i_z} &= M_{r_{ix}} \\ I_{r_{yy}} \dot{\omega}_{i_y} + I_{r_{xx}} \Omega_z \omega_{i_x} + I_{r_{xy}} \Omega_z \omega_{i_y} - I_{r_{zz}} \Omega_x \omega_{i_z} &= M_{r_{iy}} \\ I_{r_{zz}} \dot{\omega}_{i_z} - I_{r_{xx}} \Omega_y \omega_{i_x} - I_{r_{xy}} \Omega_y \omega_{i_y} + I_{r_{yy}} \Omega_x \omega_{i_x} + I_{r_{xy}} \Omega_x \omega_{i_z} &= M_{r_{iz}} \end{aligned} \quad (3)$$

The propellers are lightweight and have a small moment of inertia compared to the vehicle. Relevant precession terms are therefore those that contain the relatively large ω_{i_z} . Because the rotors spin around the z axis, it is safe to assume that $\omega_{i_x} \ll \omega_{i_z}$ and $\omega_{i_y} \ll \omega_{i_z}$ and that $\dot{\omega}_{i_x}$ and $\dot{\omega}_{i_y}$ are negligible. Then, the moments exerted on the rotors due to their rotational dynamics are given by Eq. (4). Note the presence of the term $I_{r_{zz}} \dot{\omega}_{i_z}$, which is the moment necessary to change the angular velocity of a rotor. In Sec. VI, it will be shown that this term is important

$$M_{r_i} = \begin{bmatrix} M_{r_{ix}} \\ M_{r_{iy}} \\ M_{r_{iz}} \end{bmatrix} = \begin{bmatrix} I_{r_{zz}} \Omega_y \omega_{i_z} \\ -I_{r_{zz}} \Omega_x \omega_{i_z} \\ I_{r_{zz}} \dot{\omega}_{i_z} \end{bmatrix} \quad (4)$$

This equation holds for each of the four rotors, and so the moment acting on a rotor is given a subscript i to indicate the rotor number. The total moment due to the rotational effects of the rotors is shown in Eq. (5). Because motors 1 and 3 spin in the opposite direction of rotors 2 and 4, a factor $(-1)^i$ is introduced. Because we are left with only the z component for the angular velocity of each rotor, we will omit this subscript and continue with the vector $\omega = [\omega_{1_z}, \dots, \omega_{4_z}]^T = [\omega_1, \dots, \omega_4]^T$:

$$\begin{aligned} M_r &= \sum_{i=1}^4 M_{r_i} = \sum_{i=1}^4 (-1)^{i+1} \begin{bmatrix} I_{r_{zz}} \Omega_y \omega_i \\ -I_{r_{zz}} \Omega_x \omega_i \\ I_{r_{zz}} \dot{\omega}_i \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ I_{r_{zz}} & -I_{r_{zz}} & I_{r_{zz}} & -I_{r_{zz}} \end{bmatrix} \begin{bmatrix} \dot{\omega}_1 \\ \dot{\omega}_2 \\ \dot{\omega}_3 \\ \dot{\omega}_4 \end{bmatrix} \\ &+ \begin{bmatrix} I_{r_{zz}} \Omega_y & -I_{r_{zz}} \Omega_y & I_{r_{zz}} \Omega_y & -I_{r_{zz}} \Omega_y \\ -I_{r_{zz}} \Omega_x & I_{r_{zz}} \Omega_x & -I_{r_{zz}} \Omega_x & I_{r_{zz}} \Omega_x \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \\ \omega_4 \end{bmatrix} \end{aligned} \quad (5)$$

Now consider the Euler equation [Eq. (1)] for the entire vehicle. The moments from the rotor dynamics are subtracted from the other moments, yielding

$$I_v \dot{\Omega} + \Omega \times I_v \Omega = M_c(\omega) + M_a(\Omega, v) - M_r(\omega, \dot{\omega}, \Omega) \quad (6)$$

Here, I_v is the moment of inertia matrix of the vehicle, $M_r(\omega, \dot{\omega}, \Omega)$ is the gyroscopic effect of the rotors, $M_c(\omega)$ is the control moment vector generated by the rotors, and $M_a(\Omega, v)$ is the moment vector generated by aerodynamic effects, which depends on the angular rates and the MAV velocity vector v . The control moment $M_c(\omega)$ is elaborated in Eq. (7), where k_1 is the force constant of the rotors, k_2 is the moment constant of the rotors, and b and l are defined in Fig. 1

[§]Data available online at https://github.com/EwoudSmeur/paparazzi/tree/bebop_indi_experiment [retrieved 23 November 2015].

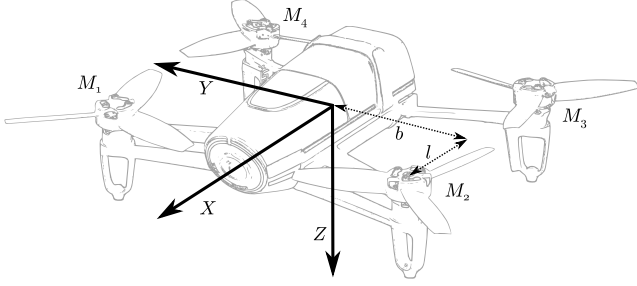


图1 实验中使用的Bebop四旋翼飞行器及轴定义。

其他研究集中于通过使用基于李雅普诺夫的控制器的设计来补偿输入的延迟 [10]。在本文中，我们展示了延迟输入（执行器动态）可以自然地由 INDI 控制器处理。

控制有效性是 INDI 仍然需要的唯一模型。参数可以通过对执行器和惯性矩的仔细建模，或者通过分析飞行日志中的输入输出数据来获得。然而，即使遵循如此繁琐的过程，控制有效性在飞行过程中仍可能发生变化。例如，这可能由于飞行条件的变化 [11] 或执行器损坏 [12] 而发生。为应对这一情况，我们提出了一种自适应确定控制有效性矩阵的方法。

在本文中，我们提出了三个主要贡献：1) 一种在数学上合理处理由陀螺仪测量滤波引起的延迟的方法，2) 引入自适应 INDI 方案，可以在线估计控制有效性，3) 在控制器设计中纳入螺旋桨动量。这些贡献在运行 Paparazzi 开源飞控软件的 Parrot Bebop 四旋翼上得到了实现和演示。这是一款商业化的四旋翼飞机，代码已在 Github 上公开。

所提出的理论和结果可以以一种直接的方式推广到其他车辆。我们已经成功地将这种控制方法应用于各种四旋翼飞行器。其中一些小型无人飞行器能够测量旋翼的旋转速度（执行器反馈），但有些则没有这种能力。人们认为，INDI 控制器能够很好地适应不同类型的小型无人飞行器，如直升机、多旋翼、固定翼或混合型飞行器。

本文的提纲如下。首先，第 II 节将讨论 MAV 的模型。其次，第 III 节将处理 INDI 及其在四旋翼控制器中的分析。第 IV 节关于 INDI 的自适应扩展。最后，第 V 节说明实验设置，第 VI 节随后给出实验结果。

II. 微型飞行器模型

如图1所示，展示了Bebop四旋翼及其坐标轴定义。执行器驱动四个旋翼，其在机体坐标系中的角速度分别为 $\omega_i = [\omega_{ix}, \omega_{iy}, \omega_{iz}]$ ，其中 i 表示旋翼编号。重心位于坐标系的原点，沿X轴到每个旋翼的距离为 l ，沿Y轴的距离为 b 。

如果车辆的角速度向量表示为 $\Omega = [p; q; r]^T$ ，其导数为 $\dot{\Omega}$ ，则旋转动力学由欧拉运动方程 [13] 给出，更具体地说，是描述旋转的方程。如果我们将机体坐标系作为我们的坐标系，则车辆的角速度为方程 (1)：

$$I_v \dot{\Omega} + \Omega \times I_v \Omega = M \quad (1)$$

其中 M 是作用在车辆上的力矩向量。如果我们考虑旋转的螺旋桨，仍然在机体坐标系中，我们得到

$$I_r \dot{\omega}_i + \Omega \times I_r \omega_i = M_{r_i} \quad (2)$$

其中 ω_i 是车辆机体坐标系中第 i 个螺旋桨的角速度向量， Ω 是坐标系的角旋转，等于车辆机体的角速度。假设旋翼在 z 轴上是平面的，因此惯性矩阵 I_r 的某些元素为零： $I_{rxz} = I_{ryz} = 0$ 。由于坐标系固定在车辆上， I_{rxx} 、 I_{ryy} 和 I_{rzz} 随时间不是常数。然而，如后面所示，包含这些惯性矩的项将消失。将公式 (2) 展开为三个分量得到

$$\begin{aligned} I_{rxx} \dot{\omega}_{ix} - I_{ryy} \Omega_z \omega_{iy} - I_{rxy} \Omega_z \omega_{ix} + I_{rzz} \Omega_y \omega_{iz} &= M_{r_{ix}} \\ I_{ryy} \dot{\omega}_{iy} + I_{rxx} \Omega_z \omega_{ix} + I_{rxy} \Omega_z \omega_{iy} - I_{rzz} \Omega_x \omega_{iz} &= M_{r_{iy}} \\ I_{rzz} \dot{\omega}_{iz} - I_{rxx} \Omega_y \omega_{ix} - I_{rxy} \Omega_y \omega_{iy} + I_{ryy} \Omega_x \omega_{ix} + I_{rxy} \Omega_x \omega_{iy} &= M_{r_{iz}} \end{aligned} \quad (3)$$

螺旋桨重量轻，与飞行器相比具有较小的转动惯量。因此，相关的进动项是那些包含相对较大 ω_{iz} 的项。由于转子围绕 z 轴旋转，可以假设 $\omega_{ix} \ll \omega_{iz}$ 和 $\omega_{iy} \ll \omega_{iz}$ 以及 $\dot{\omega}_{ix}$ 和 $\dot{\omega}_{iy}$ 可以忽略不计。然后，由于旋转动力学作用在转子上的力矩由方程 (4) 给出。注意存在项 $I_{rzz} \dot{\omega}_{iz}$ ，这是改变转子角速度所需的力矩。在第六节中，将会显示这一项是重要的。

$$M_{r_i} = \begin{bmatrix} M_{r_{ix}} \\ M_{r_{iy}} \\ M_{r_{iz}} \end{bmatrix} = \begin{bmatrix} I_{rzz} \Omega_y \omega_{iz} \\ -I_{rzz} \Omega_x \omega_{iz} \\ I_{rzz} \dot{\omega}_{iz} \end{bmatrix} \quad (4)$$

这个方程适用于四个旋翼中的每一个，因此作用在某个旋翼上的力矩以下标 i 表示旋翼的编号。由于旋翼的旋转效应产生的总力矩在公式 (5) 中给出。由于电机 1 和 3 的旋转方向与旋翼 2 和 4 相反，所以引入了一个因子 $(-1)^i$ 。由于每个旋翼的角速度只剩下 z 分量，我们将省略此下标，并继续使用向量 $\omega = [\omega_{1z}; \dots; \omega_{4z}]^T = [\omega_1; \dots; \omega_4]^T$ ：

$$\begin{aligned} M_r &= \sum_{i=1}^4 M_{r_i} = \sum_{i=1}^4 (-1)^{i+1} \begin{bmatrix} I_{rzz} \Omega_y \omega_i \\ -I_{rzz} \Omega_x \omega_i \\ I_{rzz} \dot{\omega}_i \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ I_{rzz} & -I_{rzz} & I_{rzz} & -I_{rzz} \end{bmatrix} \begin{bmatrix} \dot{\omega}_1 \\ \dot{\omega}_2 \\ \dot{\omega}_3 \\ \dot{\omega}_4 \end{bmatrix} \\ &+ \begin{bmatrix} I_{rzz} \Omega_y & -I_{rzz} \Omega_y & I_{rzz} \Omega_y & -I_{rzz} \Omega_y \\ -I_{rzz} \Omega_x & I_{rzz} \Omega_x & -I_{rzz} \Omega_x & I_{rzz} \Omega_x \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \\ \omega_4 \end{bmatrix} \end{aligned} \quad (5)$$

现在考虑整个车辆的欧拉方程 [方程 (1)]。螺旋桨动力学产生的力矩从其他力矩中被减去，得到

$$I_v \dot{\Omega} + \Omega \times I_v \Omega = M_c(\omega) + M_a(\Omega, v) - M_r(\omega, \dot{\omega}, \Omega) \quad (6)$$

这里， I_v 是车辆的转动惯量矩阵， $M_r(\omega; \dot{\omega}; \Omega)$ 是旋翼的陀螺效应， $M_c(\omega)$ 是旋翼产生的控制力矩向量，而 $M_a(\Omega; v)$ 是由空气动力学效应产生的力矩向量，其依赖于角速度和 MAV 速度向量 v 。控制力矩 $M_c(\omega)$ 在公式 (7) 中有所阐述，其中 k_1 是旋翼的力常数， k_2 是旋翼的力矩常数， b 和 l 定义如图 1 所示。

[§]Data available online at https://github.com/EwoudSmeur/paparazzi/tree/bebop_indi_experiment [retrieved 23 November 2015].

$$M_c = \begin{bmatrix} bk_1(-\omega_1^2 + \omega_2^2 + \omega_3^2 - \omega_4^2) \\ lk_1(\omega_1^2 + \omega_2^2 - \omega_3^2 - \omega_4^2) \\ k_2(\omega_1^2 - \omega_2^2 + \omega_3^2 - \omega_4^2) \\ -bk_1 \quad bk_1 \quad bk_1 \quad -bk_1 \\ lk_1 \quad lk_1 \quad -lk_1 \quad -lk_1 \\ k_2 \quad -k_2 \quad k_2 \quad -k_2 \end{bmatrix} \omega^2 \quad (7)$$

If we now take Eq. (6), insert Eqs. (4) and (7), and solve for the angular acceleration $\dot{\Omega}$, we arrive at the following:

$$\begin{aligned} \dot{\Omega} &= I_v^{-1}(M_a(\Omega, v) - \Omega \times I_v \Omega) + I_v^{-1}(M_c - M_r) \\ &= F(\Omega, v) + \frac{1}{2}G_1\omega^2 - T_s G_2 \dot{\omega} - C(\Omega)G_3\omega \end{aligned} \quad (8)$$

where $F(\Omega, v) = I_v^{-1}(M_a(\Omega, v) - \Omega \times I_v \Omega)$ are the forces independent of the actuators, and G_1 , G_2 , G_3 , and $C(\Omega)$ are given by Eqs. (9–12), respectively. Note that the sample time T_s of the quadrotor is introduced to ease future calculations:

$$G_1 = 2I_v^{-1} \begin{bmatrix} -bk_1 & bk_1 & bk_1 & -bk_1 \\ lk_1 & lk_1 & -lk_1 & -lk_1 \\ k_2 & -k_2 & k_2 & -k_2 \end{bmatrix} \quad (9)$$

$$G_2 = I_v^{-1} T_s^{-1} \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ I_{r_{zz}} & -I_{r_{zz}} & I_{r_{zz}} & -I_{r_{zz}} \end{bmatrix} \quad (10)$$

$$G_3 = I_v^{-1} \begin{bmatrix} I_{r_{zz}} & -I_{r_{zz}} & I_{r_{zz}} & -I_{r_{zz}} \\ -I_{r_{zz}} & I_{r_{zz}} & -I_{r_{zz}} & I_{r_{zz}} \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (11)$$

$$C(\Omega) = \begin{bmatrix} \Omega_y & 0 & 0 \\ 0 & \Omega_x & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (12)$$

Note that traditionally in the literature, the system solved by INDI has the form of $\dot{x} = f(x) + g(x, u)$ where x is the state of the system and u the input to the system. However, as becomes clear from Eq. (8), the quadrotor is actually a system of the form $\dot{x} = f(x) + g(x, u, \dot{u})$. In Sec. III, a solution to this type of problem will be shown.

III. Incremental Nonlinear Dynamic Inversion

Consider Eq. (8) from the previous section. This equation has some extra terms compared to previous work [8] because the gyroscopic and angular momentum effects of the rotors are included. We can apply a Taylor expansion to Eq. (8) and if we neglect higher-order terms, this results in Eq. (13):

$$\begin{aligned} \dot{\Omega} &= F(\Omega_0, v_0) + \frac{1}{2}G_1\omega_0^2 + T_s G_2 \dot{\omega}_0 - C(\Omega_0)G_3\omega_0 \\ &+ \frac{\partial}{\partial \Omega} (F(\Omega, v_0) + C(\Omega)G_3\omega_0)|_{\Omega=\Omega_0} (\Omega - \Omega_0) \\ &+ \frac{\partial}{\partial v} (F(\Omega_0, v))|_{v=v_0} (v - v_0) \\ &+ \frac{\partial}{\partial \omega} \left(\frac{1}{2}G_1\omega^2 - C(\Omega_0)G_3\omega \right) \Big|_{\omega=\omega_0} (\omega - \omega_0) \\ &+ \frac{\partial}{\partial \dot{\omega}} (T_s G_2 \dot{\omega}) \Big|_{\dot{\omega}=\dot{\omega}_0} (\dot{\omega} - \dot{\omega}_0) \end{aligned} \quad (13)$$

This equation predicts the angular acceleration after an infinitesimal time step ahead in time based on a change in angular rates of the vehicle and a change in rotational rate of the rotors. Now observe that the first terms give the angular acceleration based on the current rates and inputs: $F(\Omega_0, v_0) + \frac{1}{2}G_1\omega_0^2 + T_s G_2 \dot{\omega}_0 - C(\Omega_0)G_3\omega_0 = \dot{\Omega}_0$. This angular acceleration can be obtained by deriving it from the angular rates, which are measured with the gyroscope. In other words, these terms are replaced by a sensor measurement, which is why INDI is also referred to as sensor-based control.

The second and third term, partial to Ω and v , are assumed to be much smaller than the fourth and fifth term, partial to ω and $\dot{\omega}$. This is commonly referred to as the principle of time scale separation [14]. This assumption only holds when the actuators are sufficiently fast and have more effect compared to the change in aerodynamic and precession moments due to changes in angular rates and body speeds. These assumptions and calculation of the partial derivatives give Eq. (14):

$$\begin{aligned} \dot{\Omega} &= \dot{\Omega}_0 + G_1 \text{diag}(\omega_0)(\omega - \omega_0) + T_s G_2 (\dot{\omega} - \dot{\omega}_0) \\ &- C(\Omega_0)G_3(\omega - \omega_0) \end{aligned} \quad (14)$$

Previously, it is stated that the angular acceleration is measured by deriving it from the angular rates. In most cases, the gyroscope measurements from a MAV are noisy due to vibrations of the vehicle due to the propellers and motors. Because differentiation of a noisy signal amplifies the noise, some filtering is required. The use of a second-order filter is adopted from the literature [9], of which a transfer function in the Laplace domain is given by Eq. (15). Satisfactory results were obtained with $\omega_n = 50$ rad/s and $\zeta = 0.55$. Other low-pass filters are also possible, for instance the Butterworth filter

$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (15)$$

The result is that, instead of the current angular acceleration, a filtered and therefore delayed angular acceleration $\dot{\Omega}_f$ is measured. Because all the terms with the zero subscript in the Taylor expansion should be at the same point in time, they are all replaced with the subscript f , yielding Eq. (16). This indicates that these signals are also filtered and are therefore synchronous with the angular acceleration:

$$\begin{aligned} \dot{\Omega} &= \dot{\Omega}_f + G_1 \text{diag}(\omega_f)(\omega - \omega_f) + T_s G_2 (\dot{\omega} - \dot{\omega}_f) \\ &- C(\Omega_f)G_3(\omega - \omega_f) \end{aligned} \quad (16)$$

This equation is not yet ready to be inverted because it contains the derivative of the angular rate of the propellers. Because we are dealing with discrete signals, consider the discrete approximation of the derivative in the z domain: $\dot{\omega} = (\omega - \omega z^{-1})T_s^{-1}$, where T_s is the sample time. This is shown in Eq. (17):

$$\begin{aligned} \dot{\Omega} &= \dot{\Omega}_f + G_1 \text{diag}(\omega_f)(\omega - \omega_f) + G_2(\omega - \omega z^{-1} - \omega_f + \omega_f z^{-1}) \\ &- C(\Omega_f)G_3(\omega - \omega_f) \end{aligned} \quad (17)$$

$$M_c = \begin{bmatrix} bk_1(-\omega_1^2 + \omega_2^2 + \omega_3^2 - \omega_4^2) \\ lk_1(\omega_1^2 + \omega_2^2 - \omega_3^2 - \omega_4^2) \\ k_2(\omega_1^2 - \omega_2^2 + \omega_3^2 - \omega_4^2) \\ -bk_1 \quad bk_1 \quad bk_1 \quad -bk_1 \\ lk_1 \quad lk_1 \quad -lk_1 \quad -lk_1 \\ k_2 \quad -k_2 \quad k_2 \quad -k_2 \end{bmatrix} \omega^2 \quad (7)$$

如果我们现在取方程 (6)，代入方程 (4) 和 (7)，并求解角加速度 $\dot{\Omega}$ ，我们得到如下结果：

$$\begin{aligned} \dot{\Omega} &= I_v^{-1}(M_a(\Omega, v) - \Omega \times I_v \Omega) + I_v^{-1}(M_c - M_r) \\ &= F(\Omega, v) + \frac{1}{2}G_1\omega^2 - T_s G_2 \dot{\omega} - C(\Omega)G_3\omega \end{aligned} \quad (8)$$

其中 $F(\Omega; v) = I_v^{-1}(M_a(\Omega; v) - \Omega \times I_v \Omega)$ 是独立于执行器的力，而 G_1 、 G_2 、 G_3 和 $C(\Omega)$ 分别由公式 (9-12) 给出。注意，引入四旋翼的采样时间 T_s 以便于后续计算：

$$G_1 = 2I_v^{-1} \begin{bmatrix} -bk_1 & bk_1 & bk_1 & -bk_1 \\ lk_1 & lk_1 & -lk_1 & -lk_1 \\ k_2 & -k_2 & k_2 & -k_2 \end{bmatrix} \quad (9)$$

$$G_2 = I_v^{-1} T_s^{-1} \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ I_{r_{zz}} & -I_{r_{zz}} & I_{r_{zz}} & -I_{r_{zz}} \end{bmatrix} \quad (10)$$

$$G_3 = I_v^{-1} \begin{bmatrix} I_{r_{zz}} & -I_{r_{zz}} & I_{r_{zz}} & -I_{r_{zz}} \\ -I_{r_{zz}} & I_{r_{zz}} & -I_{r_{zz}} & I_{r_{zz}} \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (11)$$

$$C(\Omega) = \begin{bmatrix} \Omega_y & 0 & 0 \\ 0 & \Omega_x & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (12)$$

请注意，传统文献中，用 INDI 求解的系统形式为 $\dot{x} = f(x) + g(x; u)$ ，其中 x 是系统的状态， u 是系统的输入。然而，从公式 (8) 可以清楚地看出，四旋翼实际上是一个形式为 $\dot{x} = f(x) + g(x; u; \dot{u})$ 的系统。在第三节中，将展示这种类型问题的解决方案。

III. 增量非线性动态反馈

考虑前一节中的式 (8)。与之前的工作 [8] 相比，该方程多了一些额外项，因为考虑了转子的陀螺效应和角动量效应。我们可以对式 (8) 进行泰勒展开，如果忽略高阶项，就得到式 (13)：

$$\begin{aligned} \dot{\Omega} &= F(\Omega_0, v_0) + \frac{1}{2}G_1\omega_0^2 + T_s G_2 \dot{\omega}_0 - C(\Omega_0)G_3\omega_0 \\ &+ \frac{\partial}{\partial \Omega} (F(\Omega, v_0) + C(\Omega)G_3\omega_0)|_{\Omega=\Omega_0} (\Omega - \Omega_0) \\ &+ \frac{\partial}{\partial v} (F(\Omega_0, v))|_{v=v_0} (v - v_0) \\ &+ \frac{\partial}{\partial \omega} \left(\frac{1}{2}G_1\omega^2 - C(\Omega_0)G_3\omega \right) \Big|_{\omega=\omega_0} (\omega - \omega_0) \\ &+ \frac{\partial}{\partial \dot{\omega}} (T_s G_2 \dot{\omega}) \Big|_{\dot{\omega}=\dot{\omega}_0} (\dot{\omega} - \dot{\omega}_0) \end{aligned} \quad (13)$$

这个方程预测在时间上一个无穷小步骤之后的角加速度，其依据是车辆的角速度变化和转子旋转速度的变化。现在注意，第一项根据当前的速率和输入给出了角加速度： $F(\Omega_0; v_0) + \frac{1}{2}G_1\omega_0^2 + T_s G_2 \dot{\omega}_0 - C(\Omega_0)G_3\omega_0 = \dot{\Omega}_0$ 。这个角加速度可以通过从角速度导出获得，而角速度是通过陀螺仪测量的。换句话说，这些项被传感器测量替代，这就是为什么 INDI 也被称为基于传感器的控制。

第二项和第三项，对 Ω 和 v 的偏导数，假设远小于第四项和第五项，对 ω 和 $\dot{\omega}$ 的偏导数。这通常被称为时间尺度分离原理 [14]。只有在执行器足够快速，并且相对于角速度和机体速度变化引起的气动力矩和进动矩变化具有更大影响时，该假设才成立。这些假设以及偏导数的计算给出了方程 (14)：

$$\begin{aligned} \dot{\Omega} &= \dot{\Omega}_0 + G_1 \text{diag}(\omega_0)(\omega - \omega_0) + T_s G_2(\dot{\omega} - \dot{\omega}_0) \\ &- C(\Omega_0)G_3(\omega - \omega_0) \end{aligned} \quad (14)$$

之前提到，角加速度是通过从角速率中导出而测量的。在大多数情况下，由于 MAV 的螺旋桨和电机带来的振动，陀螺仪测量值存在噪声。由于对噪声信号进行微分会放大噪声，因此需要进行一些滤波。文献 [9] 采用了二阶滤波器，其在拉普拉斯域中的传递函数如公式 (15) 所示。使用 $\omega_n = 50 \text{ rad/s}$ 和 $\zeta = 0.55$ 得到了令人满意的结果。其他低通滤波器也可使用，例如巴特沃斯滤波器。

$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (15)$$

其结果是，不是测量当前的角加速度，而是测量经过滤波因此延迟的角加速度 $\dot{\Omega}_f$ 。由于泰勒展开中所有带有零下标的项应在相同的时间点，因此它们都被替换为下标 f ，从而得到公式 (16)。这表明这些信号也经过滤波，因此与角加速度同步：

$$\begin{aligned} \dot{\Omega} &= \dot{\Omega}_f + G_1 \text{diag}(\omega_f)(\omega - \omega_f) + T_s G_2(\dot{\omega} - \dot{\omega}_f) \\ &- C(\Omega_f)G_3(\omega - \omega_f) \end{aligned} \quad (16)$$

该方程尚未准备好进行求逆，因为它包含了螺旋桨角速度的导数。由于我们处理的是离散信号，因此考虑 z 域中导数的离散近似： $\dot{\omega} = (\omega - \omega z^{-1})T_s^{-1}$ ，其中 T_s 是采样时间。这在方程 (17) 中展示：

$$\begin{aligned} \dot{\Omega} &= \dot{\Omega}_f + G_1 \text{diag}(\omega_f)(\omega - \omega_f) + G_2(\omega - \omega z^{-1} - \omega_f + \omega_f z^{-1}) \\ &- C(\Omega_f)G_3(\omega - \omega_f) \end{aligned} \quad (17)$$

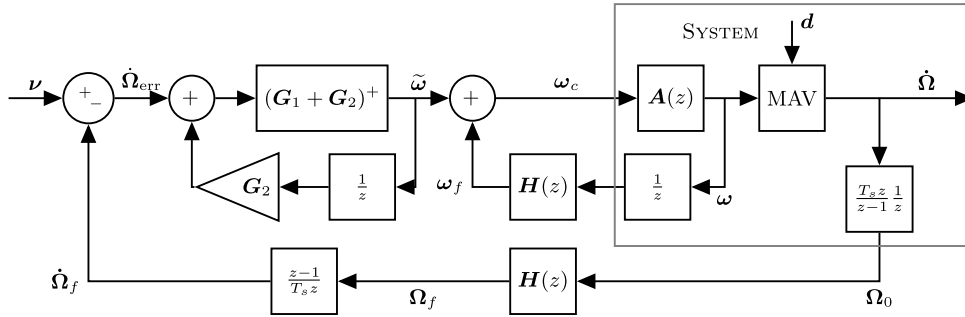


Fig. 2 INDI control scheme. $A(z)$ denotes the actuator dynamics, and $H(z)$ is the second-order filter.

Collecting all terms with $(\omega - \omega_f)$ yields Eq. (18):

$$\dot{\Omega} = \dot{\Omega}_f + (G_1 \text{diag}(\omega_f) + G_2 - C(\Omega_f)G_3)(\omega - \omega_f) - G_2 z^{-1}(\omega - \omega_f) \quad (18)$$

Inversion of this equation for ω yields Eq. (19), where $^+$ denotes the Moore–Penrose pseudoinverse:

$$\omega_c = \omega_f + (G_1 \text{diag}(\omega_f) + G_2 - C(\Omega_f)G_3)^+ (\nu - \dot{\Omega}_f + G_2 z^{-1}(\omega_c - \omega_f)) \quad (19)$$

Note that the predicted angular acceleration $\dot{\Omega}$ is now instead a virtual control, denoted by ν . The virtual control is the desired angular acceleration, and with Eq. (19), the required inputs ω_c can be calculated. The subscript c is added to ω to indicate that this is the command sent to the motors. This input is given with respect to a previous input ω_f . If we define the increment in the motor commands as $\tilde{\omega} = \omega_c - \omega_f$, it is clearly an incremental control law.

A. Parameter Estimation

Equation (19) shows the general quadrotor INDI control law. The parameters of this equation are the three matrices G_1 , G_2 , and G_3 , which need to be identified for the specific quadrotor. This can be done through measurement of each of the components that make up these matrices, including the moments of inertia of the vehicle and the propellers as well as the thrust and drag coefficients of the rotors. Identifying the parameters in this way requires a significant amount of effort.

A more effective method is to use test flight data to determine the model coefficients. Of course, to do this, the MAV needs to be flying. This can be achieved by initially tuning the parameters. Alternatively, a different controller can be used at first to gather the test flight data, such as PID control. Once a test flight has been logged, Eq. (18) is used for parameter estimation and is written as Eq. (20). From this equation, a least-squares solution is found for the matrices G_1 , G_2 , and G_3

$$\Delta \dot{\Omega}_f = [G_1 \quad G_2 \quad C(\Omega_f)G_3] \begin{bmatrix} \text{diag}(\omega_f)\Delta\omega_f \\ (\Delta\omega_f - z^{-1}\Delta\omega_f) \\ -\Delta\omega_f \end{bmatrix} \quad (20)$$

Here, Δ denotes the finite difference between two subsequent samples. From the data, we can also investigate the importance of some of the terms by comparing the least-squares error with and without the terms. It turns out that, on a typical dataset, leaving out the matrix G_3 only results in an estimation squared error increase of $\sim 0.2\%$. Furthermore, modeling the rotor as linear with the rotational speed of the rotor instead of quadratic gives an estimation squared error increase of $\sim 0.9\%$. Therefore, we can simplify the INDI control law of Eq. (19) into Eq. (21):

$$\omega_c = \omega_f + (G_1 + G_2)^+ (\nu - \dot{\Omega}_f + G_2 z^{-1}(\omega_c - \omega_f)) \quad (21)$$

B. Implementation

With the simplifications described in Sec. III.A, the final INDI control scheme is shown in Fig. 2. The input to the system is the virtual control ν , and the output is the angular acceleration of the system, $\dot{\Omega}$. The angular velocity measurement from the gyroscope is fed back through the differentiating second-order filter and subtracted from the virtual control to give the angular acceleration error $\dot{\Omega}_{\text{err}}$.

Because the matrices G_1 and G_2 are not square, we take the pseudoinverse to solve the problem of control allocation, denoted by $^+$. The contents of the block “MAV” are shown in Fig. 3 because it allows the closed-loop analysis in Sec. III.C. In this diagram, d is a disturbance term that bundles disturbances and unmodeled dynamics.

Note that Eq. (21) provides a desired angular velocity of the rotors. However, the actuators do not have an instantaneous response. Instead, it is assumed they have first-order dynamics $A(z)$. The reference sent to the motors is denoted by ω_c and $\tilde{\omega} = \omega_c - \omega_f$. In Fig. 2, it is assumed that actuator feedback is available. However, if this is not the case, the actuator state ω_0 has to be estimated with a model of the actuator dynamics as is shown in Fig. 4. Here, $A'(z)$ is a model of the actuator dynamics.

C. Closed-Loop Analysis

Consider the control diagram shown in Fig. 2. We can verify that this is a stable controller by doing a closed-loop analysis. First, the transfer function of each of the two small loops is calculated, shown by Eqs. (22) and (23). Here, $TF_{x \rightarrow y}$ denotes the transfer function from point x to y in the control diagram:

$$\begin{aligned} \tilde{\omega} &= (G_1 + G_2)^+ \dot{\Omega}_{\text{err}} + (G_1 + G_2)^+ G_2 z^{-1} \tilde{\omega} \\ (G_1 + G_2) \tilde{\omega} &= \dot{\Omega}_{\text{err}} + G_2 z^{-1} \tilde{\omega} \\ (G_1 + G_2 - G_2 z^{-1}) \tilde{\omega} &= \dot{\Omega}_{\text{err}} \\ TF_{\dot{\Omega}_{\text{err}} \rightarrow \tilde{\omega}}(z) &= (G_1 + G_2 - G_2 z^{-1})^+ \end{aligned} \quad (22)$$

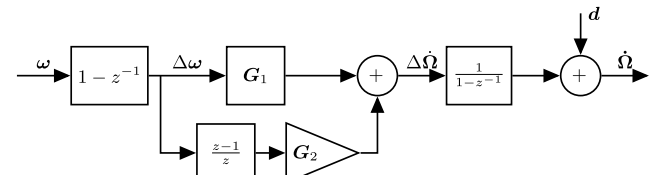


Fig. 3 Contents of the block named “MAV” in Fig. 2.

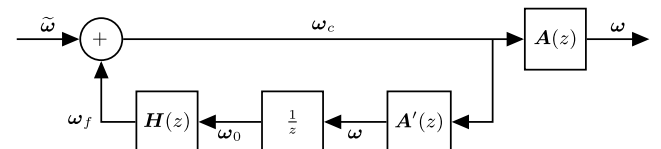


Fig. 4 Block diagram for estimation of actuator state if actuator feedback is not available.

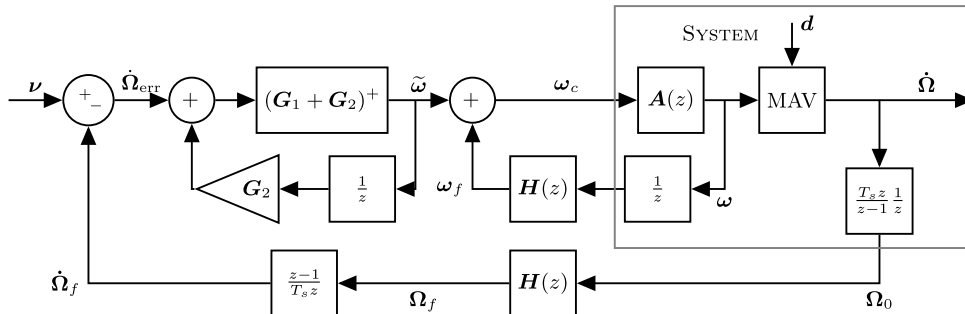


图2 INDI 控制方案。A(z) 表示执行器动力学，H(z) 是二阶滤波器。

将所有含有 $(\omega - \omega_f)$ 的项收集起来得到方程 (18): “}

$$\begin{aligned} \dot{\Omega} = \dot{\Omega}_f + (G_1 \text{diag}(\omega_f) + G_2 - C(\Omega_f)G_3)(\omega - \omega_f) \\ - G_2 z^{-1}(\omega - \omega_f) \end{aligned} \quad (18)$$

将该方程对 ω 求逆得到方程 (19)，其中 $+$ 表示 Moore-Penrose 伪逆：

$$\begin{aligned} \omega_c = \omega_f + (G_1 \text{diag}(\omega_f) + G_2 - C(\Omega_f)G_3)^+ \\ (\nu - \dot{\Omega}_f + G_2 z^{-1}(\omega_c - \omega_f)) \end{aligned} \quad (19)$$

请注意，预测的角加速度 $\dot{\Omega}$ 现在变为一个虚拟控制，记为 ν 。虚拟控制是期望的角加速度，并且通过公式 (19)，所需的输入 ω_c 可以被计算出来。下标 c 被加到 ω 以表示这是发送给电机的指令。这个输入是相对于先前的输入 ω_f 给出的。如果我们将电机指令的增量定义为 $\tilde{\omega} = \omega_c - \omega_f$ ，显然它是一个增量控制律。

A. 参数估计

方程 (19) 显示了一般四旋翼 INDI 控制律。该方程的参数是三个矩阵 G_1 、 G_2 和 G_3 ，需要针对特定四旋翼进行识别。这可以通过测量组成这些矩阵的各个分量来完成，包括飞行器和螺旋桨的转动惯量以及旋翼的推力和阻力系数。以这种方式识别参数需要大量的工作。

一种更有效的方法是使用试飞数据来确定模型系数。当然，要做到这一点，MAV 需要处于飞行状态。这可以通过初始调整参数来实现。或者，也可以首先使用不同的控制器来收集试飞数据，例如 PID 控制。一旦记录了试飞数据，就使用方程 (18) 进行参数估计，并写成方程 (20)。通过该方程，可以找到矩阵 G_1 、 G_2 和 G_3 的最小二乘解。

$$\Delta \dot{\Omega}_f = [G_1 \quad G_2 \quad C(\Omega_f)G_3] \begin{bmatrix} \text{diag}(\omega_f)\Delta\omega_f \\ (\Delta\omega_f - z^{-1}\Delta\omega_f) \\ -\Delta\omega_f \end{bmatrix} \quad (20)$$

这里， Δ 表示两个连续样本之间的有限差分。根据数据，我们还可以通过比较有无某些项的最小二乘误差来研究这些项的重要性。结果显示，在典型数据集中，省略矩阵 G_3 仅会导致估计平方误差增加 $\sim 0.2\%$ 。此外，将转子建模为随转子旋转速度线性变化而不是二次变化，会使估计平方误差增加 $\sim 0.9\%$ 。因此，我们可以将方程 (19) 的 INDI 控制律简化为方程 (21)：

$$\omega_c = \omega_f + (G_1 + G_2)^+(\nu - \dot{\Omega}_f + G_2 z^{-1}(\omega_c - \omega_f)) \quad (21)$$

B. 实施

根据第 III.A 节中描述的简化，最终的 INDI 控制方案如图 2 所示。系统的输入是虚拟控制 ν ，输出是系统的角加速度 $\dot{\Omega}$ 。来自陀螺仪的角速度测量通过二阶微分滤波器反馈，并从虚拟控制中减去，从而得到角加速度误差 $\dot{\Omega}_{\text{err}}$ 。

因为矩阵 G_1 和 G_2 不是方阵，我们取伪逆来解决控制分配问题，记为 $+$ 。“MAV” 模块的内容如图 3 所示，因为它允许在第 III.C 节中进行闭环分析。在该图中， d 是一个扰动项，包含扰动和未建模的动力学。

注意，方程 (21) 提供了转子的期望角速度。然而，执行器没有瞬时响应。相反，假设它们具有一阶动力学 $A(z)$ 。发送给电机的参考信号记为 ω_c 和 $\tilde{\omega} = \omega_c - \omega_f$ 。在图 2 中，假设可以获得执行器反馈。然而，如果情况并非如此，则必须使用执行器动力学模型来估计执行器状态 ω_0 ，如图 4 所示。这里， $A'(z)$ 是执行器动力学的模型。

C. 闭环分析

考虑图 2 所示的控制图。我们可以通过进行闭环分析来验证这是一个稳定的控制器。首先，计算两个小环路的传递函数，如公式 (22) 和 (23) 所示。这里， $TF_{x \rightarrow y}$ 表示控制图中从点 x 到 y 的传递函数：

$$\begin{aligned} \tilde{\omega} &= (G_1 + G_2)^+ \dot{\Omega}_{\text{err}} + (G_1 + G_2)^+ G_2 z^{-1} \tilde{\omega} \\ (G_1 + G_2) \tilde{\omega} &= \dot{\Omega}_{\text{err}} + G_2 z^{-1} \tilde{\omega} \\ (G_1 + G_2 - G_2 z^{-1}) \tilde{\omega} &= \dot{\Omega}_{\text{err}} \\ TF_{\dot{\Omega}_{\text{err}} \rightarrow \tilde{\omega}}(z) &= (G_1 + G_2 - G_2 z^{-1})^+ \end{aligned} \quad (22)$$

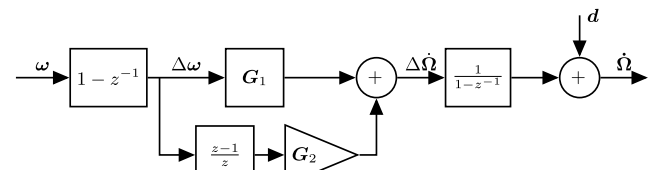


图3 图2中名为“MAV”的模块内容。

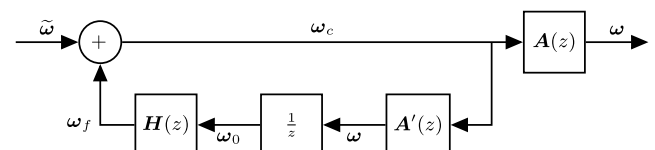


图4 如果无法获得执行器反馈时的执行器状态估计框图。

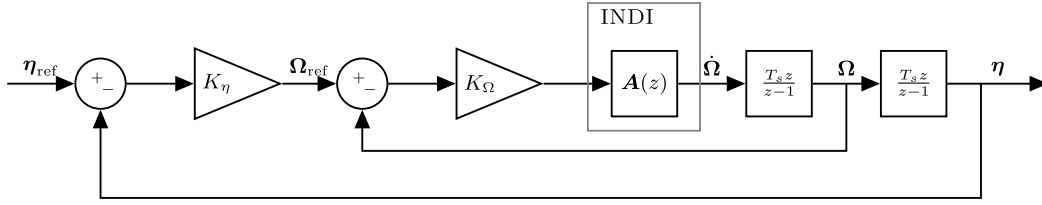


Fig. 5 Design of the attitude controller based on the closed-loop response of the INDI controller.

We define $H(z) = IH(z)$ and assume that all actuators have the same dynamics, and so $A(z) = IA(z)$. This means that each matrix in $TF_{\ddot{\omega} \rightarrow \omega}(z)$ is a diagonal matrix, and therefore $TF_{\ddot{\omega} \rightarrow \omega}(z)$ is a diagonal matrix function:

$$\begin{aligned} TF_{\ddot{\omega} \rightarrow \omega}(z) &= (I - A(z)H(z)z^{-1})^{-1}A(z) \\ &= (I - IA(z)IH(z)z^{-1})^{-1}IA(z) \\ &= (I(1 - A(z)H(z)z^{-1}))^{-1}IA(z) \\ &= I(1 - A(z)H(z)z^{-1})^{-1}A(z) \end{aligned} \quad (23)$$

Then, the last part of the open loop is from ω to $\dot{\Omega}$, as shown by Fig. 3. Using this figure, the transfer function is calculated in Eq. (24). Note that, for this analysis, disturbances are not taken into account:

$$TF_{\omega \rightarrow \dot{\Omega}}(z) = G_1 + \frac{z-1}{z}G_2 = G_1 + G_2 - G_2z^{-1} \quad (24)$$

Using these intermediate results, the open-loop transfer function of the entire system is shown in Eq. (25):

$$\begin{aligned} TF_{\dot{\Omega}_{\text{err}} \rightarrow \dot{\Omega}}(z) &= TF_{\omega \rightarrow \dot{\Omega}}(z)TF_{\ddot{\omega} \rightarrow \omega}(z)TF_{\dot{\Omega}_{\text{err}} \rightarrow \ddot{\omega}}(z) \\ &= (G_1 + G_2 - G_2z^{-1})I(1 - A(z)H(z)z^{-1})^{-1}A(z) \\ &\quad (G_1 + G_2 - G_2z^{-1})^+ \\ &= I(1 - A(z)H(z)z^{-1})^{-1}A(z) \end{aligned} \quad (25)$$

Using Eq. (25) and Fig. 2, we can calculate the closed-loop transfer function of the entire system in Eq. (26):

$$\begin{aligned} TF_{\nu \rightarrow \dot{\Omega}}(z) &= (I + TF_{\dot{\Omega}_{\text{err}} \rightarrow \dot{\Omega}}(z)IH(z)z^{-1})^{-1}TF_{\dot{\Omega}_{\text{err}} \rightarrow \dot{\Omega}}(z) \\ &= (I + I(1 - A(z)H(z)z^{-1})^{-1}A(z)IH(z)z^{-1})^{-1} \\ &\quad I(1 - A(z)H(z)z^{-1})^{-1}A(z) \\ &= I \frac{(1 - A(z)H(z)z^{-1})^{-1}A(z)}{1 + (1 - A(z)H(z)z^{-1})^{-1}A(z)H(z)z^{-1}} \\ &= I \frac{A(z)}{1 - A(z)H(z)z^{-1} + A(z)H(z)z^{-1}} \\ &= IA(z) \end{aligned} \quad (26)$$

From this equation, it appears that the closed-loop transfer function from the virtual input to the angular acceleration is, in fact, the actuator dynamics $A(z)$. In most cases, the actuator dynamics can be represented by first- or second-order dynamics. Note that this shows the importance of applying the $H(z)$ filter on the input as well. By doing this, a lot of terms cancel, and all that remains is the actuator dynamics.

Now, consider the transfer function from disturbances d (see Fig. 2) to the angular acceleration. The derivation is given in Eq. (27) in which use is made of Eq. (25):

$$\begin{aligned} TF_{d \rightarrow \dot{\Omega}}(z) &= (I - TF_{\dot{\Omega}_{\text{err}} \rightarrow \dot{\Omega}}(z)(-1)H(z)z^{-1})^{-1}I \\ &= (I + I(1 - A(z)H(z)z^{-1})^{-1}A(z)IH(z)z^{-1})^{-1}I \\ &= I \frac{1}{1 + (1 - A(z)H(z)z^{-1})^{-1}A(z)H(z)z^{-1}} \\ &= I \frac{1 - A(z)H(z)z^{-1}}{1 - A(z)H(z)z^{-1} + A(z)H(z)z^{-1}} \\ &= I(1 - A(z)H(z)z^{-1}) \end{aligned} \quad (27)$$

With Eq. (27), we show that disturbances in the angular acceleration are rejected as long as the actuator dynamics and the designed filter are stable. The term $A(z)H(z)z^{-1}$ will go to 1 over time, with a response determined by the actuator dynamics, filter dynamics, and a unit delay. This means that if the angular acceleration is measured faster, the drone can respond to disturbances faster. Moreover, if the actuators can react faster, disturbances can be neutralized faster.

D. Attitude Control

The angular acceleration of the MAV is accurately controlled by the system shown in Fig. 2. To control the attitude of the MAV, a stabilizing angular acceleration reference needs to be passed to the INDI controller. This outer-loop controller can be as simple as a proportional derivative (PD) controller (a gain on the rate error and a gain on the angle error), as shown in Fig. 5. Here, η represents the attitude of the quadcopter. The benefit of the INDI inner-loop controller is that the outer PD controller commands a reference, independent of the effectiveness of the actuators (including the inertia of the quadrotor).

This means that the design of this controller depends only on the speed of the actuator dynamics $A(z)$. In case the actuator dynamics are known (through analysis of logged test flights, for instance), values of K_η and K_Ω can be determined that give a stable response.

This outer-loop controller does not involve inversion of the attitude kinematics, as has been done in other work [3]. However, the attitude angles for a quadrotor are generally small, in which case the inversion of the attitude kinematics can be replaced with simple angle feedback.

E. Altitude Control

The INDI controller derived in the beginning of this section controls the angular acceleration around the axes x , y , and z , which correspond to roll, pitch, and yaw. However, there is a fourth degree of freedom that is controlled with the rotors, which is the acceleration along the z axis.

Control of this fourth axis is handled by a separate controller. This controller scales the average input to the motors to a value commanded by the pilot, after the input has been incremented by the INDI controller.

IV. Adaptive Incremental Nonlinear Dynamic Inversion

The INDI approach only relies on modeling of the actuators. The control effectiveness depends on the moment of inertia of the vehicle as well as the type of motors and propellers. A change in any of these will require re-estimation of the control effectiveness. Moreover, the

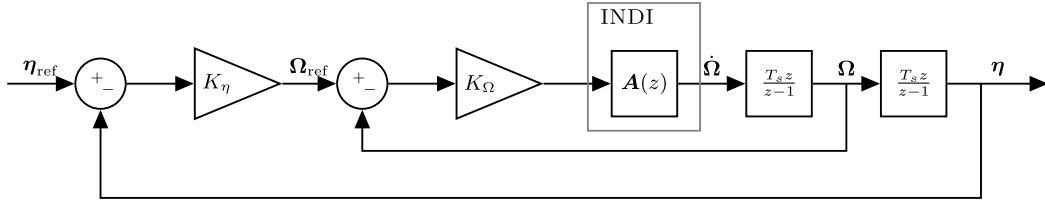


图5 基于INDI控制器闭环响应的姿态控制器设计。

我们定义 $H(z) = IH(z)$ 并假设所有执行器具有相同的动力学，因此 $A(z) = IA(z)$ 。这意味着 $TF_{\dot{\omega} \rightarrow \omega}(z)$ 中的每个矩阵都是对角矩阵，因此 $TF_{\dot{\omega} \rightarrow \omega}(z)$ 是一个对角矩阵函数：

$$\begin{aligned} TF_{\dot{\omega} \rightarrow \omega}(z) &= (I - A(z)H(z)z^{-1})^{-1}A(z) \\ &= (I - IA(z)IH(z)z^{-1})^{-1}IA(z) \\ &= (I(1 - A(z)H(z)z^{-1}))^{-1}IA(z) \\ &= I(1 - A(z)H(z)z^{-1})^{-1}A(z) \end{aligned} \quad (23)$$

然后，开环的最后部分是从 ω 到 Ω ，如图 3 所示。利用该图，传递函数在式 (24) 中计算。请注意，对于此分析，不考虑干扰：

$$TF_{\omega \rightarrow \Omega}(z) = G_1 + \frac{z-1}{z}G_2 = G_1 + G_2 - G_2z^{-1} \quad (24)$$

利用这些中间结果，整个系统的开环传递函数如公式 (25) 所示：

$$\begin{aligned} TF_{\dot{\Omega}_{err} \rightarrow \dot{\Omega}}(z) &= TF_{\omega \rightarrow \Omega}(z)TF_{\dot{\omega} \rightarrow \omega}(z)TF_{\dot{\Omega}_{err} \rightarrow \dot{\omega}}(z) \\ &= (G_1 + G_2 - G_2z^{-1})I(1 - A(z)H(z)z^{-1})^{-1}A(z) \\ &\quad (G_1 + G_2 - G_2z^{-1})^+ \\ &= I(1 - A(z)H(z)z^{-1})^{-1}A(z) \end{aligned} \quad (25)$$

使用公式(25)和图2，我们可以计算出整个系统的闭环传递函数，如公式(26)所示：

$$\begin{aligned} TF_{\nu \rightarrow \dot{\Omega}}(z) &= (I + TF_{\dot{\Omega}_{err} \rightarrow \dot{\Omega}}(z)IH(z)z^{-1})^{-1}TF_{\dot{\Omega}_{err} \rightarrow \dot{\Omega}}(z) \\ &= (I + I(1 - A(z)H(z)z^{-1})^{-1}A(z)IH(z)z^{-1})^{-1} \\ &\quad I(1 - A(z)H(z)z^{-1})^{-1}A(z) \\ &= I \frac{(1 - A(z)H(z)z^{-1})^{-1}A(z)}{1 + (1 - A(z)H(z)z^{-1})^{-1}A(z)H(z)z^{-1}} \\ &= I \frac{A(z)}{1 - A(z)H(z)z^{-1} + A(z)H(z)z^{-1}} \\ &= IA(z) \end{aligned} \quad (26)$$

从这个方程来看，从虚拟输入到角加速度的闭环传递函数实际上是执行器动力学 $A(z)$ 。在大多数情况下，执行器动力学可以用一阶或二阶动力学来表示。需要注意的是，这也显示了在输入上应用 $H(z)$ 滤波器的重要性。通过这样做，许多项会相互抵消，最终只剩下执行器动力学。

现在，考虑从扰动 d （见图 2）到角加速度的传递函数。推导在公式 (27) 中给出，其中使用了公式 (25)：

$$\begin{aligned} TF_{d \rightarrow \dot{\Omega}}(z) &= (I - TF_{\dot{\Omega}_{err} \rightarrow \dot{\Omega}}(z)(-1)H(z)z^{-1})^{-1}I \\ &= (I + I(1 - A(z)H(z)z^{-1})^{-1}A(z)IH(z)z^{-1})^{-1}I \\ &= I \frac{1}{1 + (1 - A(z)H(z)z^{-1})^{-1}A(z)H(z)z^{-1}} \\ &= I \frac{1 - A(z)H(z)z^{-1}}{1 - A(z)H(z)z^{-1} + A(z)H(z)z^{-1}} \\ &= I(1 - A(z)H(z)z^{-1}) \end{aligned} \quad (27)$$

通过方程 (27)，我们表明，只要执行器动力学和设计的滤波器是稳定的，角加速度中的扰动就会被抑制。项 $A(z)H(z)z^{-1}$ 随时间会趋于 1，其响应由执行器动力学、滤波器动力学以及单位延迟决定。这意味着，如果角加速度测量得更快，无人机可以更快地响应扰动。此外，如果执行器能够更快地反应，扰动可以被更快地抵消。

D. 姿态控制

如图 2 所示，系统可以精确地控制微型航空器的角加速度。为了控制微型航空器的姿态，需要向 INDI 控制器传递一个稳定的角加速度参考值。这个外环控制器可以像图 5 所示那样简单地采用比例微分 (PD) 控制器（对速率误差的增益和对角度误差的增益）。其中， η 表示四旋翼的姿态。INDI 内环控制器的好处在于，外环 PD 控制器发出的参考指令与执行器的有效性（包括四旋翼的惯性）无关。

这意味着该控制器的设计仅取决于执行器动力学 $A(z)$ 的速度。如果执行器动力学已知（例如通过对记录的试飞数据进行分析），可以确定 K_η 和 K_Ω 的值，从而获得稳定的响应。

这个外环控制器不涉及姿态运动学的反演，如其他工作 [3] 所做的那样。然而，对于四旋翼来说，姿态角通常很小，在这种情况下，姿态运动学的反演可以用简单的角度反馈替代。

E. 高度控制

本节开头推导的 INDI 控制器控制绕 x 、 y 和 z 轴的角加速度，这些轴对应滚转、俯仰和偏航。然而，还有第四个自由度是通过旋翼控制的，即沿 z 轴的加速度。

第四轴的控制由单独的控制器处理。该控制器在输入经过 INDI 控制器增量调整后，将输入的平均值按飞行员指令缩放到电机所需的数值。

四、自适应增量非线性动态反演

INDI 方法仅依赖于执行器的建模。控制效能取决于飞行器的转动惯量以及电机和螺旋桨的类型。任何这些参数的变化都需要重新估算控制效能。此外，

control effectiveness can even change during flight, due to a change in flight velocity, battery voltage, or actuator failure.

To counteract these problems and obtain a controller that requires no manual parameter estimation, the controller was extended with onboard adaptive parameter estimation using a least mean squares (LMS) [15] adaptive filter. This filter is often used in adaptive signal filtering and adaptive neural networks.

The LMS implementation is shown in Eq. (28), where μ_1 is a diagonal matrix whose elements are the adaptation constant for each input, and μ_2 is a diagonal matrix to adjust the adaptation constants per axis. This is necessary because not all axes have the same signal-to-noise ratio.

The LMS formula calculates the difference between the expected acceleration based on the inputs and the measured acceleration. Then, it increments the control effectiveness based on the error. The control effectiveness includes both G_1 and G_2 , as is shown in Eq. (29):

$$G(k) = G(k-1) - \mu_2 \left(G(k-1) \begin{bmatrix} \Delta\omega_f \\ \Delta\dot{\omega}_f \end{bmatrix} - \Delta\dot{\Omega}_f \right) \begin{bmatrix} \Delta\omega_f \\ \Delta\dot{\omega}_f \end{bmatrix}^T \mu_1 \quad (28)$$

$$G = [G_1 \quad G_2] \quad (29)$$

Clearly, when there is no change in input, the control effectiveness is not changed. The reverse is also true; more excitation of the system will result in a faster adaptation. This is a benefit of the LMS algorithm over, for instance, recursive least squares with a finite horizon because recursive least squares will “forget” everything outside the horizon. Note that the filtering for the online parameter estimation can be different from the filtering for the actual control. Equation (28) makes use of $\Delta\dot{\Omega}_f$, which is the finite difference of $\dot{\Omega}_f$ in the control Eq. (21). Because differentiating amplifies high frequencies, a filter that provides more attenuation of these high frequencies is necessary. We still use the second-order filter described by Eq. (15), but with $\omega_n = 25$ rad/s and $\zeta = 0.55$.

When an approximate control effectiveness is given before takeoff, the adaptive system will estimate the actual values online and thereby tune itself. The only knowledge provided to the controller is an initial guess of the control effectiveness. It is generally not possible to take off without any estimate of the control effectiveness because the UAV might crash before the adaptive system has converged.

The choice of the adaptation constants μ_1 and μ_2 determines the stability and the rate of adaptation. By making these constants larger, a faster convergence is achieved. By making them too large, the adaptation will no longer be stable. The theoretical limit has been discussed in the literature [15], and it depends on the autocorrelation matrix of the input to the filter. In practice, the filter stability deteriorates before the theoretical limit, and so to find a good adaptation constant, some tuning is required.

V. Experimental Setup

To validate the performance of the INDI controller developed in Sec. III and the adaptive parameter estimation from Sec. IV, several experiments were conducted. These experiments were performed using the Bebop quadcopter from Parrot shown in Fig. 1. The Bebop weighs 396.2 g and can be equipped with bumpers, which are 12 g per bumper. For these experiments, the bumpers were not equipped unless explicitly stated. The quadcopter was running the Paparazzi open-source autopilot software, which contains all the code for wireless communication, reading sensor measurements, etc. The accelerometer, gyroscope, and control loops were running at 512 Hz.

Four experiments test the key properties of the controller: 1) performance, 2) disturbance rejection, and 3) adaptation.

During these experiments, the reference attitude and average thrust level were controlled by a pilot and sent to the drone over Wi-Fi. All other computations were done on the drone itself, including the online adaptation.

A. Performance

To put the responsiveness of the system to the test and to make sure that the angular acceleration reference is tracked by the INDI controller, a doublet input was applied on the attitude roll angle. The amplitude of the doublet is 30 deg, and the period is half a second (0.25 s positive and 0.25 s negative). This test is only done for the roll and not for the pitch because there is no fundamental difference between these axes. The yaw axis is covered separately in Sec. V.D. Note that this experiment is performed without the adaptation.

The performance is compared to a manually tuned PID controller. The INDI controller is not expected to be faster or slower than a traditional PID controller because the result of Eq. (26) shows that the response of the INDI inner loop is simply the actuator dynamics. Considering that the outer loop is a PD controller, the rise time and overshoot should be similar.

Finally, this test will also be performed with an INDI controller that does not contain the filter delay compensation, more specifically by using ω_0 in the controller increment instead of ω_f . It is expected that this will not fly well, because in Sec. III.C, we showed that with this compensation all terms cancel, and the closed-loop transfer function reduces to $IA(z)$.

By inspection of Fig. 2, we can get a feel for what will happen if we omit this filter compensation. When there is an angular acceleration error, a control increment $\tilde{\omega}$ will be the result, which is added to ω_0 to produce ω_c . ω_c goes through the actuator dynamics to produce the new ω . The next time step, the result of this new ω , does not yet appear in $\dot{\Omega}_f$, because it is filtered and therefore delayed. Therefore, $\tilde{\omega}$ will be the same. However, ω_0 did update, and so ω_c will be incremented even more, while we are still waiting to see the result of the first increment in $\dot{\Omega}_f$.

B. Disturbance Rejection

The disturbance rejection property is validated by adding a disturbance to the system. One possibility would be to apply aerodynamic disturbances by flying in the wake of a big fan. The disturbances occurring would be realistic but not very repeatable. Moreover, the magnitude of the disturbance would be unknown.

Instead, it is possible to apply a disturbance in the form of a step function to the system. This is done by adding a weight of 42.5 g to a container located in an off-centered position on the quadrotor while it is flying, as shown in Fig. 6. The container is located on the front of the drone and has a distance of about 11 cm to the center of gravity, and so any weight added will shift the center of gravity forward. This will cause a misalignment of the thrust vector with respect to the center of gravity and therefore a pitch moment. This moment will be persistent and therefore have the form of a step disturbance. This is indicated with d in Fig. 2. Although this moment is created with a center of gravity shift, the situation is the same as in the case of a persistent gust or an unmodeled aerodynamic moment.

A normal PID controller would respond to such a disturbance very slowly because it takes time for the integrator to accumulate. But the introduction of the INDI inner loop leads to a cascaded control structure, which is much more resistant to disturbances than a single-



Fig. 6 Container attached to the nose of the quadrotor with one weight inside.

由于飞行速度、电池电压或执行器故障的变化，控制效能甚至可能在飞行过程中发生变化。

为了应对这些问题并获得一个不需要手动参数估计的控制器，该控制器通过使用最小均方 (LMS) [15] 自适应滤波器的车载自适应参数估计进行了扩展。这种滤波器通常用于自适应信号滤波和自适应神经网络。

LMS 实现如公式 (28) 所示，其中 μ_1 是一个对角矩阵，其元素是每个输入的自适应常数，而 μ_2 是一个对角矩阵，用于调整每个轴的自适应常数。这是必要的，因为并非所有轴的信噪比都相同。

LMS 公式计算基于输入的期望加速度与测量加速度之间的差异。然后，它根据误差增加控制有效性。控制有效性包括 G_1 和 G_2 ，如公式 (29) 所示：

$$G(k) = G(k-1) - \mu_2 \left(G(k-1) \begin{bmatrix} \Delta \omega_f \\ \Delta \dot{\omega}_f \end{bmatrix} - \Delta \dot{\Omega}_f \right) \begin{bmatrix} \Delta \omega_f \\ \Delta \dot{\omega}_f \end{bmatrix}^T \mu_1 \quad (28)$$

$$G = [G_1 \quad G_2] \quad (29)$$

显然，当输入没有变化时，控制有效性不会改变。反之亦然；系统的激励越多，适应速度越快。这是 LMS 算法相对于例如具有有限视野的递归最小二乘法的一个优势，因为递归最小二乘法会“遗忘”视野之外的所有内容。请注意，用于在线参数估计的滤波可以与实际控制的滤波不同。方程 (28) 使用了 $\Delta \Omega_f$ ，它是控制方程 (21) 中 Ω_f 的有限差分。因为微分会放大高频，需要一个能够更多衰减这些高频的滤波器。我们仍然使用方程 (15) 描述的二阶滤波器，但 $\omega_n = 25 \text{ rad/s}$ ， $\zeta = 0.55$ 。

当在起飞前给出大致的控制效能时，自适应系统将在线估计实际值，从而调整自身。提供给控制器的唯一信息是控制效能的初步猜测。通常在没有任何控制效能估计的情况下无法起飞，因为无人机可能在自适应系统收敛之前坠毁。

适应常数 μ_1 和 μ_2 的选择决定了适应的稳定性和速度。将这些常数调大，可以实现更快的收敛；但如果过大，适应将不再稳定。文献 [15] 中讨论了理论极限，其取决于滤波器输入的自相关矩阵。在实际中，滤波器的稳定性在达到理论极限之前就会恶化，因此为了找到合适的适应常数，需要进行一定的调节。

五、实验设置

为了验证在第 III 节中开发的 INDI 控制器的性能以及第 IV 节中的自适应参数估计，进行了若干实验。这些实验使用了图 1 所示的 Parrot Bebop 四旋翼无人机进行。Bebop 重量为 396.2 克，可以装备缓冲器，每个缓冲器重 12 克。在这些实验中，除非明确说明，否则未安装缓冲器。四旋翼无人机运行 Paparazzi 开源自动驾驶软件，其中包含用于无线通信、读取传感器测量等的全部代码。加速度计、陀螺仪和控制回路的运行频率为 512 Hz。四个实验测试了控制器的关键特性：1) 性能，2) 扰动抑制，3) 自适应能力。

在这些实验中，参考姿态和平均推力水平由飞行员控制，并通过 Wi-Fi 发送到无人机。所有其他计算都在无人机自身完成，包括在线适应。

A. 性能

为了测试系统的响应性，并确保 INDI 控制器能够跟踪角加速度参考值，在姿态滚转角上施加了双脉冲输入。双脉冲的幅值为 30 度，周期为半秒（0.25 秒为正，0.25 秒为负）。此测试仅在滚转角上进行，而不在俯仰角上进行，因为这些轴之间没有根本区别。偏航轴在第 V.D 节另行讨论。请注意，该实验是在未进行自适应的情况下进行的。

性能与手动调节的 PID 控制器进行比较。预计 INDI 控制器不会比传统的 PID 控制器快或慢，因为方程 (26) 的结果显示 INDI 内环的响应仅仅是执行器的动态。考虑到外环是 PD 控制器，上升时间和超调量应该相似。

最后，这个测试也将使用不包含滤波器延迟补偿的 INDI 控制器来进行，更具体地说，是通过在控制器增量中使用 ω_0 而不是 ω_f 。预计效果不会很好，因为在第 III.C 节中，我们已经展示了有了该补偿后所有项都会相互抵消，闭环传递函数简化为 $1A(z)$ 。

通过检查图 2，我们可以感受到如果我们省略这个滤波补偿会发生什么。当存在角加速度误差时，将产生一个控制增量 $\tilde{\omega}$ ，该增量加到 ω_0 上以产生 ω_c 。 ω_c 经过执行器动态产生新的 ω 。下一时间步，这个新的 ω 的结果尚未出现在 Ω_f 中，因为它被滤波，因此会有延迟。因此， $\tilde{\omega}$ 将保持不变。然而， ω_0 已经更新，因此 ω_c 会进一步增加，而我们仍在等待第一个增量在 Ω_f 中的结果。

B. 扰动抑制

通过向系统添加扰动来验证扰动抑制特性。一种可能的方式是通过飞行在大风扇的尾流中来施加气动扰动。发生的扰动将是现实的，但不太可重复。此外，扰动的大小将是未知的。

相反，可以向系统施加以阶跃函数形式的扰动。这是通过在四旋翼飞行时，在偏置位置的容器上增加 42.5 克的重量来实现的，如图 6 所示。该容器位于无人机的前部，距离重心约 11 厘米，因此添加的任何重量都会将重心向前移动。这将导致推力矢量与重心不对齐，从而产生俯仰力矩。该力矩将是持续存在的，因此呈现阶跃扰动的形式。在图 2 中用 d 表示。尽管该力矩是通过重心移动产生的，但情况与持续气流扰动或未建模的空气动力矩的情况相同。

一个普通的 PID 控制器对这种扰动的响应会非常慢，因为积分器积累需要时间。但引入 INDI 内环导致了级联控制结构，这比单一控制器对扰动的抵抗力要强得多。



图6 四旋翼机鼻部安装的容器，里面放有一个重量。



Fig. 7 Bebop quadrotor with bumpers.

loop design [16]. Because of this, the reference pitch angle is expected to be tracked shortly after the disturbance.

C. Adaptation

The Bebop quadcopter has the possibility to fly with bumpers, as is shown in Fig. 7. Though these bumpers only weigh 12 g apiece, they are located far from the center of gravity and therefore increase the moment of inertia. Furthermore, they can influence the airflow around the propellers. These system changes affect the G_1 and G_2 matrices. Therefore, the adaptive algorithm from Sec. IV should deal with adding or removing the bumpers.

First, two flights are performed to show the effect of adding or removing the bumpers when the adaptive algorithm is not active. For the first flight, the bumpers are added, whereas the G_1 and G_2 matrices correspond to the quadrotor without bumpers. For the second flight, the bumpers are removed, and the G matrices from the quadrotor with bumpers are used. In both flights, doublets are performed like in Sec. V.A. The performance is expected to degrade compared to the previous results for both cases because the G matrices do not correspond to what they should be.

Second, the ability of the quadrotor to adapt its G_1 and G_2 matrices is tested. In this experiment, the drone starts with bumpers equipped, but with system matrices that represent the configuration without bumpers. The pilot flies the drone in a confined area while performing some pitch, roll, and yaw maneuvers to excite the system. While flying, the correct matrices should be estimated. Then, the Bebop is landed, and the bumpers are removed. After takeoff, the matrices should converge to their original state.

Finally, doublets are performed with and without the bumpers equipped while the adaptation algorithm is active. We expect the same performance as in Sec. V.A.

D. Yaw Control

The purpose of this experiment is to show the improvement in yaw performance due to the incorporation of the rotor spin-up torque in the controller design. This is done by applying a doublet input on the yaw set point. The amplitude of the doublet is 5 deg, and the period is 1 s (0.5 s positive and 0.5 s negative). As a comparison, the same experiment is performed with a traditional PID controller. This PID controller is manually tuned to give a fast rise time with minimal overshoot.

Additionally, the same test is performed with a zero G_2 matrix. Here, we expect an oscillation because the persistent effect of a change in rotor angular velocity on the yaw axis is small. We take the pseudoinverse in Eq. (21), and so the resulting gain will be very large. Because there is the angular momentum effect of the propellers, the initial angular acceleration will be larger than expected, and the controller will start to oscillate.

VI. Results

This section deals with the results of the experiments described in Sec. V. The angular acceleration shown in the plots in this section is not the onboard estimate of the angular acceleration because it is delayed through filtering. Instead, it is computed after the experiment from the finite difference of the gyroscope data. The signal is filtered with a fourth-order Butterworth filter with a cutoff frequency of 15 Hz. It is filtered twice (forward and reverse), resulting in a zero-phase (noncausal) filter. For the actual control, the onboard filtered (and delayed) angular acceleration was used.

A. Performance

Figure 8 shows the angular acceleration around the x axis denoted by $\ddot{p}$ and the reference angular acceleration denoted by $\ddot{p}_{\text{ref}}$. Additionally, the reference is filtered with the actuator dynamics, resulting in $\ddot{p}_{\text{ref}_A}$. This signal is the angular acceleration that is expected based on the calculations in Sec. III.C, specifically Eq. (26). It might seem that the controller does not track the reference well because it lags behind the reference, but this was expected based on the model of the actuator dynamics. The angular acceleration is actually very close to the expected angular acceleration $\ddot{p}_{\text{ref}_A}$. Finally, we also show the angular acceleration as calculated onboard the quadrotor using the second-order filter. The filtered angular acceleration onboard the quadrotor is significantly delayed with respect to the actual angular acceleration, which is why we will run into problems if we do not take this delay into account in the INDI controller.

The outer-loop controller, which generates the angular acceleration reference to track, was designed such that the resultant accelerations give a desired response of the roll angle, shown in Fig. 9. From this figure, it can be seen that the quadcopter reaches its reference roll angle within 0.2 s with a very small overshoot.

The roll angle response of the PID controller is shown in Fig. 10. As expected, the PID controller performs very similar to the INDI controller in terms of rise time and overshoot. The integral gain included in the PID controller, which needs to eliminate steady-state offsets, degrades the dynamic performance of the closed-loop system. This shows that the INDI controller marginally improves the

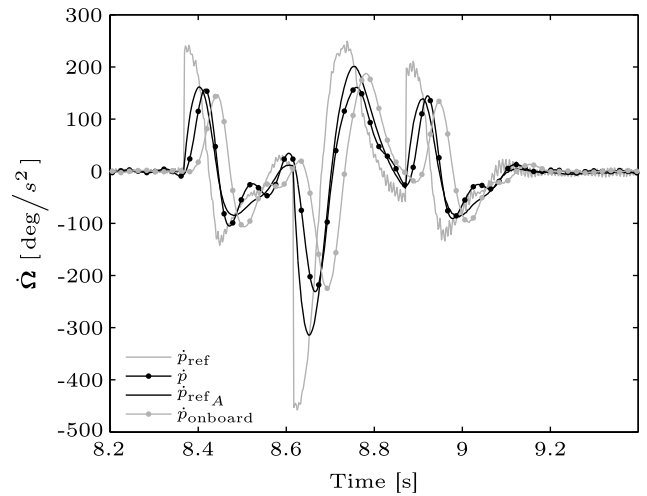


Fig. 8 Angular acceleration in the roll axis during doublet input.

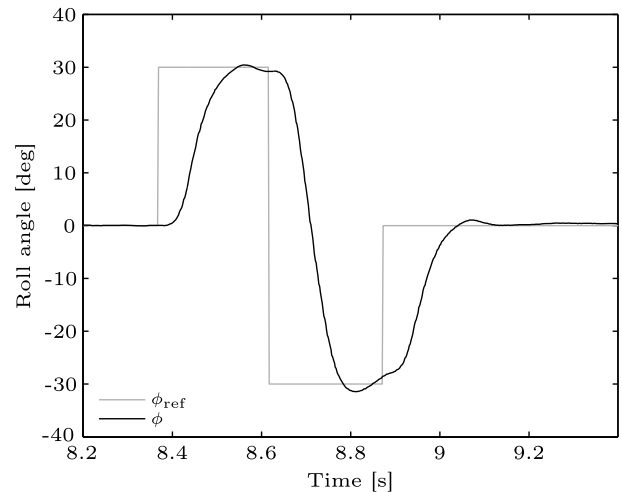


Fig. 9 Roll angle during the doublet for the INDI controller.



图7 带防撞条的Bebop四旋翼飞行器。

回路设计 [16]。因此，参考俯仰角预计将在扰动发生后不久被跟踪。

C. 适应

Bebop 四旋翼无人机可以安装防撞杆飞行，如图 7 所示。虽然这些防撞杆每个仅重 12 克，但它们位于重心的远端，因此会增加转动惯量。此外，它们还可能影响螺旋桨周围的气流。这些系统变化会影响 G_1 和 G_2 矩阵。因此，第 IV 节中的自适应算法应处理防撞杆的添加或移除。

首先，进行了两次飞行，以显示在自适应算法未激活时添加或移除缓冲器的效果。在第一次飞行中，安装了缓冲器，而 G_1 和 G_2 矩阵对应的是没有缓冲器的四旋翼。第二次飞行中，移除了缓冲器，并使用带缓冲器的四旋翼的 G 矩阵。在两次飞行中，都像第 V.A 节那样进行双升降操作。与之前的结果相比，这两种情况下的性能预计都会下降，因为 G 矩阵与应有的情况不符。

其次，测试了四旋翼调整其 G_1 和 G_2 矩阵的能力。在此实验中，无人机初始时安装了防撞条，但系统矩阵表示的是未安装防撞条的配置。飞行员在一个有限空间内操作无人机，同时进行一些俯仰、横滚和偏航动作以激发系统。在飞行过程中，应估算出正确的矩阵。然后，Bebop 着陆并移除防撞条。起飞后，矩阵应收敛回原始状态。

最后，在适应算法激活的情况下，双工实验在配备和未配备缓冲器的条件下进行。我们期望与第五章 A 节中的表现相同。

D. 偏航控制

本实验的目的是展示在控制器设计中引入旋翼加速扭矩后，偏航性能的改善。这是通过在偏航设定点上施加双脉冲输入来实现的。双脉冲的幅值为 5 度，周期为 1 秒（正向 0.5 秒，反向 0.5 秒）。作为对比，同样的实验也在传统 PID 控制器下进行。该 PID 控制器经过手动调节，以实现快速上升时间和最小超调量。

此外，同样的测试在零 G_2 矩阵下进行。在这里，我们预期会出现振荡，因为转子角速度变化对偏航轴的持续影响很小。我们在公式 (21) 中取伪逆，因此得到的增益会非常大。由于螺旋桨的角动量效应，初始角加速度将比预期的大，控制器将开始振荡。

六、结果

本节处理第 V 节中描述实验的结果。本节图中显示的角加速度并不是机载的角加速度估计值，因为它经过滤波会有延迟。相反，它是通过实验后陀螺仪数据的有限差分计算得出的。信号通过一个截止频率为 15 Hz 的四阶巴特沃斯滤波器进行滤波。信号被滤波两次（正向和反向），从而得到零相位（非因果）滤波器。在实际控制中，使用的是机载滤波（并延迟后的）角加速度。

A. 性能

图 8 显示了绕 x 轴的角加速度，用 $\dot{p}$ 表示，以及参考角加速度，用 $\dot{p}_{ref}$ 表示。此外，参考值经过执行器动力学滤波，得到 $\dot{p}_{ref,A}$ 。该信号是基于第 III.C 节的计算（特别是式 (26)）所期望的角加速度。可能看起来控制器没有很好地跟踪参考，因为它落后于参考，但根据执行器动力学模型，这是预期的。角加速度实际上与预期的角加速度 $\dot{p}_{ref,A}$ 非常接近。最后，我们还展示了无人机上使用二阶滤波计算的角加速度。无人机上滤波后的角加速度相对于实际角加速度明显滞后，这就是为什么如果在 INDI 控制器中不考虑这一延迟，我们会遇到问题的原因。

外环控制器产生用于跟踪的角加速度参考，其设计使得产生的加速度能够给出滚转角的期望响应，如图 9 所示。从该图可以看出，四旋翼在 0.2 秒内达到其参考滚转角，并且超调非常小。

PID 控制器的横滚角响应如图 10 所示。如预期的那样，PID 控制器在上升时间和超调方面的表现与 INDI 控制器非常相似。PID 控制器中包含的积分增益需要消除稳态偏差，但会降低闭环系统的动态性能。这表明 INDI 控制器在一定程度上改进了

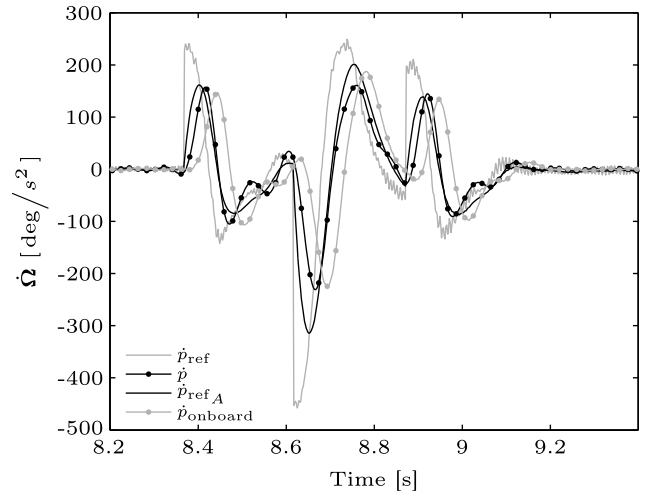


图8 双重输入下横滚轴的角加速度。

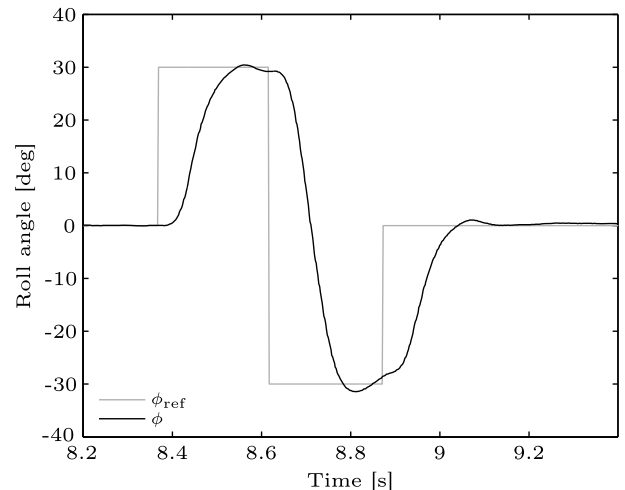


图9 INDI控制器在双重脉冲中的横滚角。

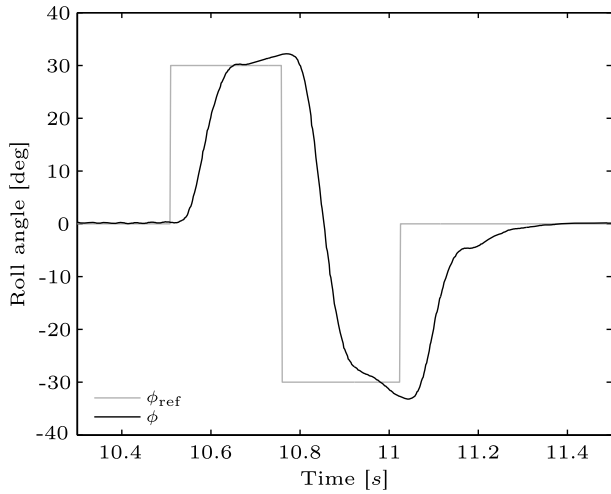


Fig. 10 Roll angle during the doublet for the PID controller.

performance of a traditional PID controller in terms of responsiveness for the roll.

As discussed previously, the onboard filtered measurement of the angular acceleration is significantly delayed. If we remove the filter delay compensation from the INDI controller, the quadrotor was severely oscillating, as can be seen in Fig. 11. The doublet was not performed because this did not seem safe. The oscillation might be reduced by lowering K_η and K_Ω , but this will make the response slower as well. From this figure, we can conclude that the filter delay compensation is an important part of the INDI controller and is crucial in obtaining good performance with an INDI controller.

B. Disturbance Rejection

The weight, shown in Fig. 6, was placed in the container attached to the nose of the quadrotor by hand. The weight was placed in the container gently, but it probably arrived in the container with some small velocity. The disturbance in the angular acceleration is therefore a combination of a step and a delta pulse.

Figure 12 shows the angular acceleration that is the result of the disturbance. From the figure, it is clear that the disturbance happened just after 13 s. As the angular acceleration increases in the negative direction, the reference angular acceleration starts to go the opposite way, because now an angular rate and a pitch angle error start to arise. About 0.1 s after losing track of the reference, the angular acceleration again coincides with the expected angular acceleration, having overcome the disturbance in the angular acceleration.

This results in a pitch angle with no steady-state error, as can be seen from Fig. 13. After 0.3 s, the pitch angle is back at zero. To show that the weight in the container really is a step disturbance, which can

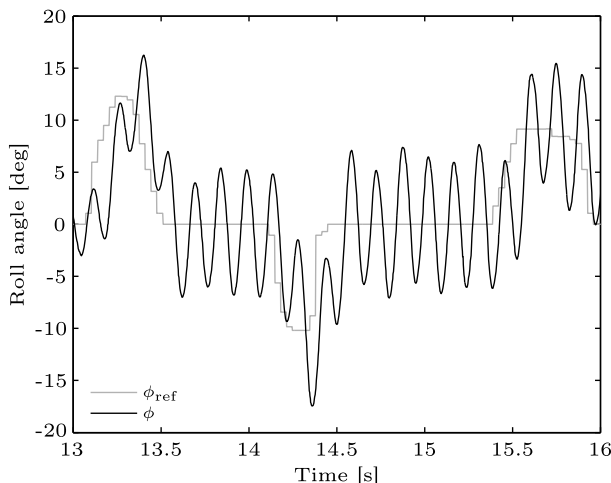


Fig. 11 Roll angle for the INDI controller without filter compensation.

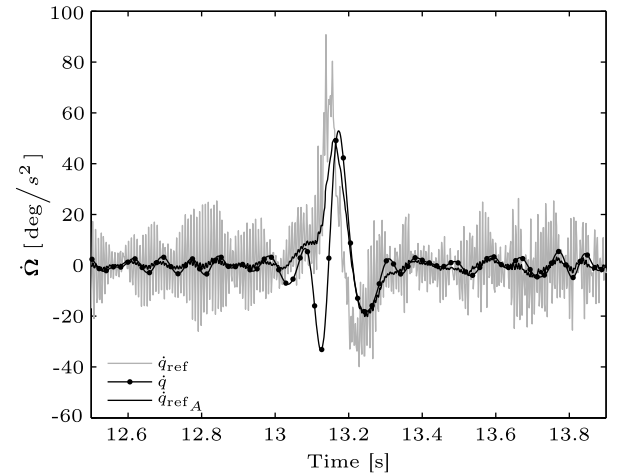


Fig. 12 Angular acceleration during the disturbance.

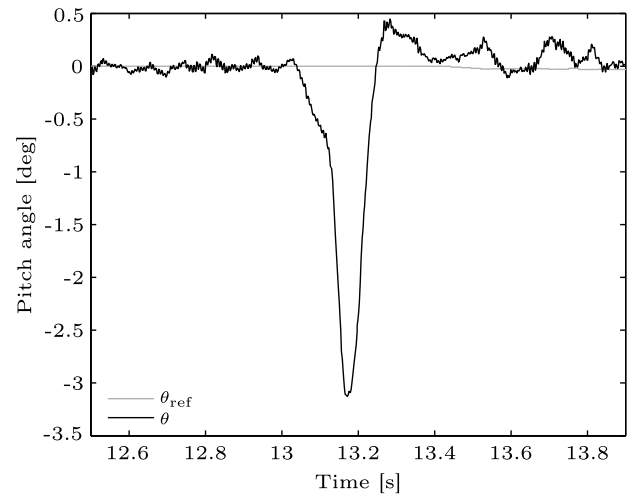


Fig. 13 Pitch angle during the disturbance.

be compared to a constant aerodynamic moment, consider Fig. 14. It shows the difference of the rotational rate of the front and rear motors divided by 4: $(\omega_1 + \omega_2 - \omega_3 - \omega_4)/4$. This indicates the average magnitude in rounds per minute that each motor contributes to the pitch control; see Eq. (7). Clearly, there is a difference before and after the disturbance, which can be quantified as an average change of 578 rounds per minute over the interval [12.6 13.0] versus [13.4 13.8].

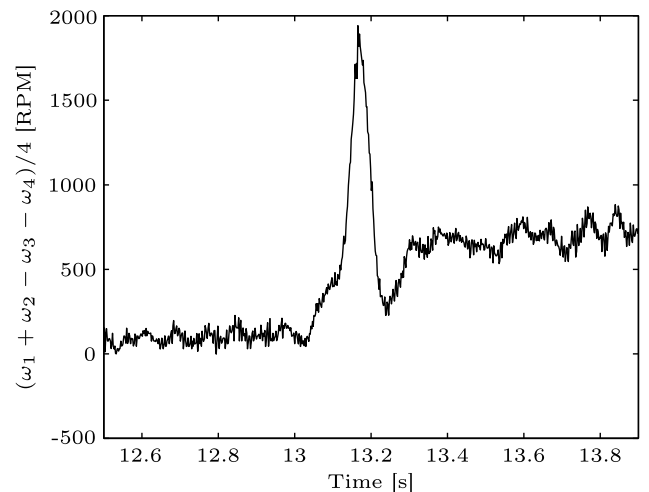


Fig. 14 Difference between the rotational rate of the front motors and the rear motors.

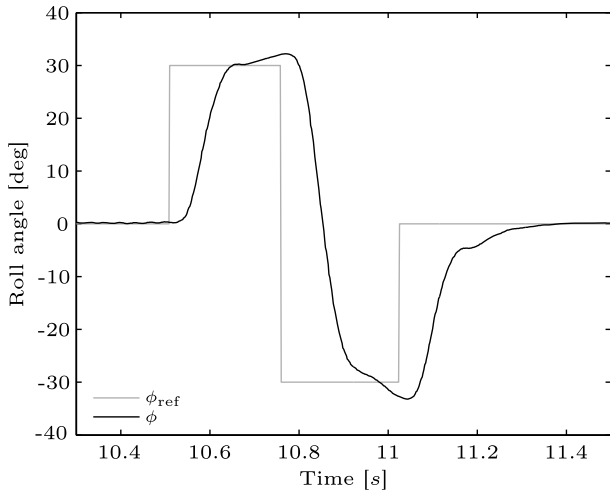


图10 PID控制器在双重扰动下的横滚角。

传统 PID 控制器在横滚响应性方面的性能。

如前所述，机载角加速度的滤波测量存在显著延迟。如果我们从 INDI 控制器中移除滤波延迟补偿，如图 11 所示，四旋翼会严重振荡。未执行双脉冲测试，因为这似乎不安全。通过降低 K_η 和 K_Ω 可能会减少振荡，但这也会使响应变慢。从该图可以看出，滤波延迟补偿是 INDI 控制器的的重要组成部分，对于获得 INDI 控制器的良好性能至关重要。

B. 干扰抑制

如图6所示，重量被手动放置在固定在四旋翼机鼻部的容器中。重量是轻轻放入容器的，但它可能还是以一定的小速度到达容器中。因此，角加速度的扰动是阶跃信号和冲击脉冲的组合。

图12显示了由扰动引起的角加速度。从图中可以清楚地看到，扰动发生在13秒之后。随着角加速度向负方向增加，参考角加速度开始向相反方向变化，因为此时角速度和俯仰角误差开始出现。在偏离参考大约0.1秒后，角加速度再次与预期角加速度一致，已经克服了角加速度中的扰动。

这会导致俯仰角没有稳态误差，正如图13所示。0.3秒后，俯仰角恢复到零。为了证明容器中的重量确实是阶跃扰动，可以

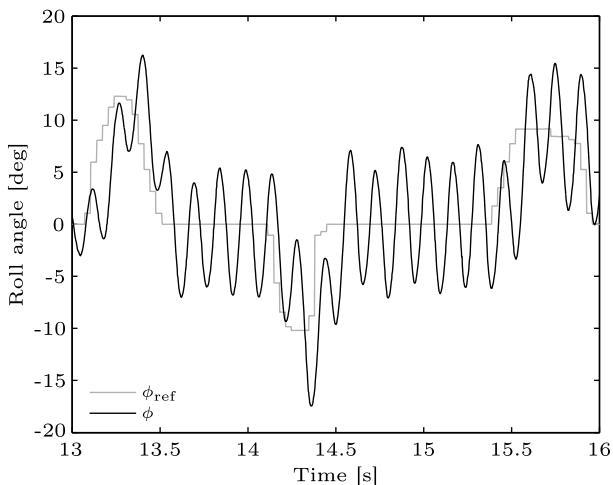


图11 INDI 控制器在没有滤波补偿下的滚转角。

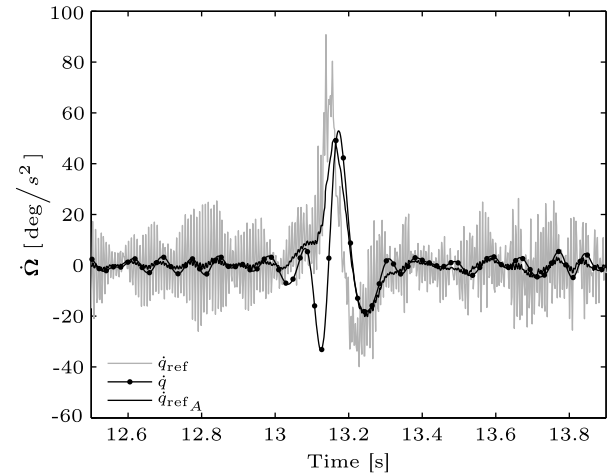


图12 扰动期间的角加速度。

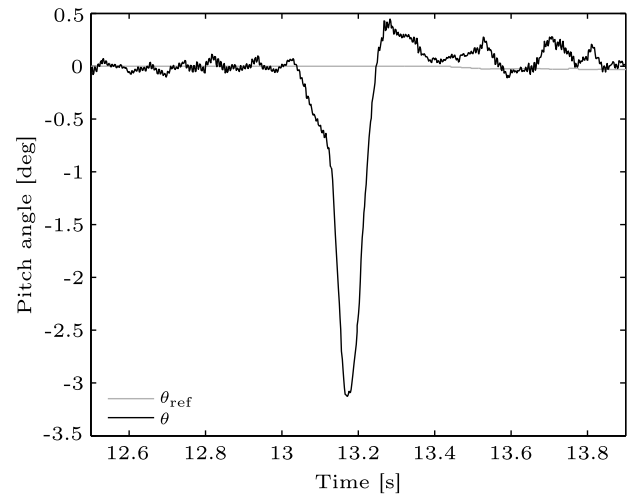


Fig. 13 Pitch angle during the disturbance.

与恒定气动转矩相比，考虑图14。它显示了前后电机转速差除以4： $(\omega_1 + \omega_2 - \omega_3 - \omega_4)/4$ 。这表示每个电机对俯仰控制的平均贡献转速（每分钟转数）；见公式(7)。显然，在扰动前后存在差异，可以量化为在区间[12.6 13.0]与[13.4 13.8]之间的平均变化量为每分钟578转。

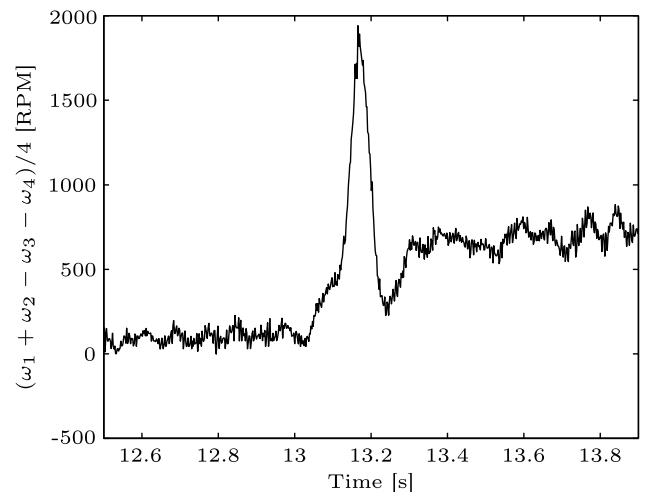


图14 前电机与后电机转速的差异。

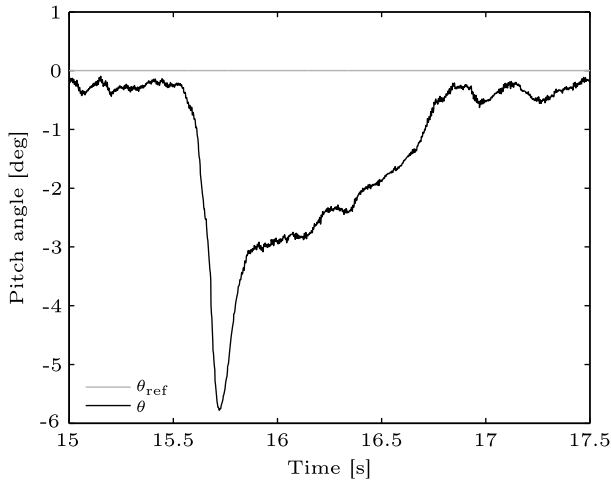


Fig. 15 Pitch angle during the disturbance for the PID controller.

This demonstrates that the disturbance was really a step and that the INDI controller can rapidly cope with such a disturbance.

Figure 15 shows the same experiment performed with a PID controller. Of course, the weight was not dropped in exactly the same manner and with the same velocity, and so the initial disturbance was probably different. However, the persisting disturbance is the same because the weight has exactly the same mass. It takes about 1.5 s before the pitch angle is back at zero again, which is approximately five times longer than for the INDI controller. One might say that the integral gain of the PID controller should be larger, but this will deteriorate the performance in the previous experiment.

C. Adaptation

Figures 16 and 17 show the response to a roll doublet without adaptation if there is a mismatch in the control effectiveness. Even though the bumpers are lightweight, their effect is significant because they are located far from the center of gravity. In Fig. 16, we see what happens if the actuators are less effective than in the model because the inertia is higher. Additional increments of the input are needed to reach a desired angular acceleration. The oscillation occurs because this takes more time. The oscillation can be reduced by reducing the K_η and K_ω gains, at the cost of having a slower response.

In Fig. 17, we see the opposite; the control effectiveness is higher than what was modeled. This results in a fast oscillation, which cannot be removed by reducing the attitude gains. This is because the cause of the oscillation is different; now, too much input is applied to reach a certain angular acceleration. This will happen regardless of what angular acceleration is requested by the attitude controller.

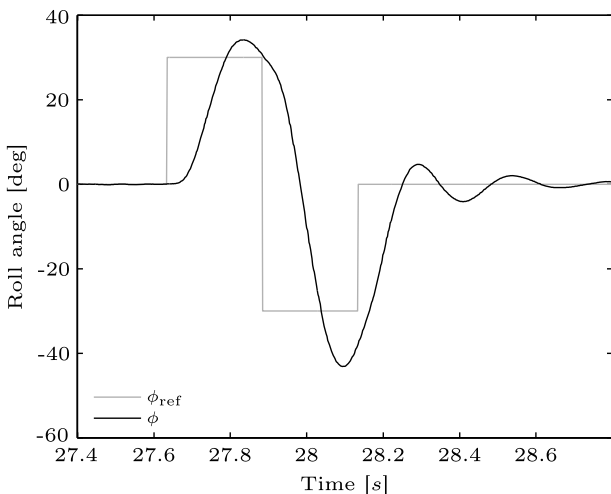


Fig. 16 Flight without adaptation, with bumpers equipped, while the control effectiveness has been determined without bumpers.

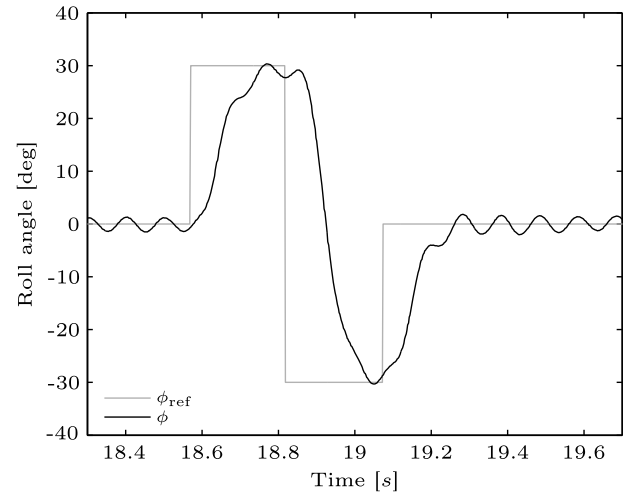


Fig. 17 Flight without adaptation, without bumpers equipped, while the control effectiveness has been determined with bumpers.

We can conclude that the performance degrades when the modeled control effectiveness does not closely correspond to the actual control effectiveness. When the adaptation algorithm is enabled, Figs. 18–20 show how each row of the G_1 matrix evolves over time as a result of the second experiment described in Sec. V.C. The same is shown in Fig. 21 for the third row of the G_2 matrix. Each line represents one of

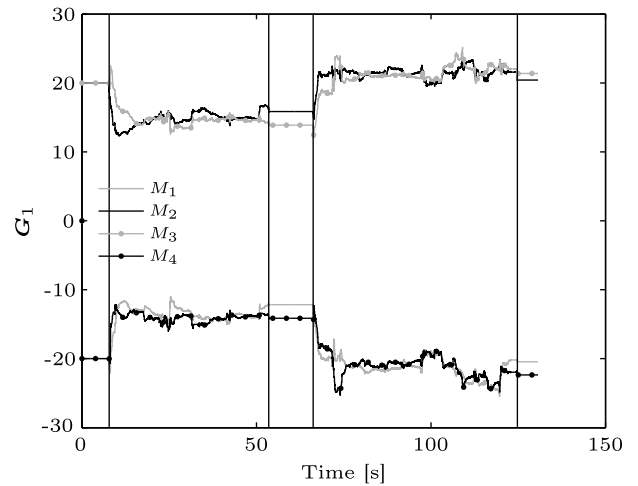


Fig. 18 First row of the G_1 matrix corresponding to the roll.

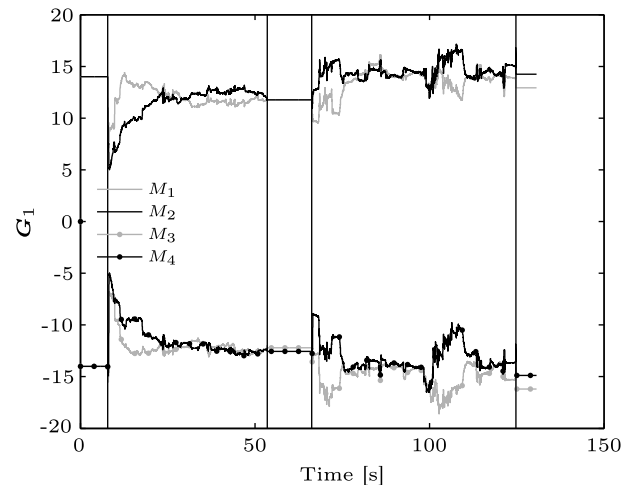


Fig. 19 Second row of the G_1 matrix corresponding to the pitch.

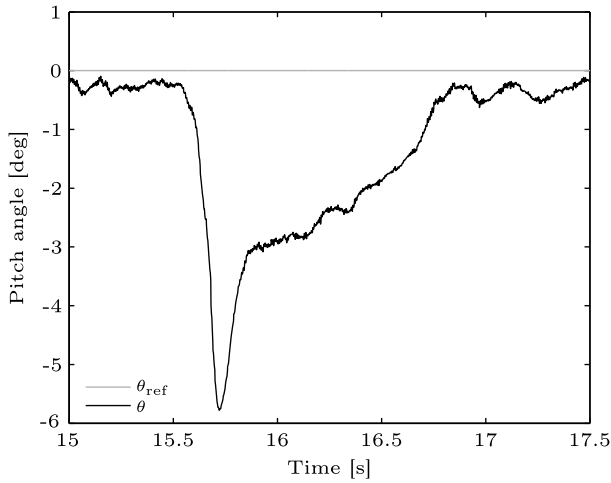


图15 PID控制器在扰动期间的俯仰角。

这表明扰动确实是一个阶跃，而且INDI控制器能够迅速应对这样的扰动。

图15显示了使用PID控制器进行的相同实验。当然，重量并不是以完全相同的方式和相同的速度下落的，因此初始扰动可能不同。然而，持续的扰动是相同的，因为重量的质量完全相同。俯仰角恢复到零大约需要1.5秒，这大约是INDI控制器所需时间的五倍。有人可能会说PID控制器的积分增益应该更大，但这会在先前的实验中降低性能。

C. 适应

图16和图17显示了在控制效能不匹配的情况下，对滚转双步响应的无适应反应。尽管缓冲器重量很轻，但由于它们位于远离重心的位置，其影响是显著的。在图16中，我们可以看到如果执行器的效能低于模型中设定的情况会发生什么，因为惯性更大。需要额外增加输入以达到期望的角加速度。振荡发生是因为这需要更多时间。可以通过降低 K_η 和 K_ω 增益来减小振荡，但代价是响应速度变慢。

在图17中，我们看到相反的情况；控制效果比建模的更高。这导致了快速振荡，通过减小姿态增益无法消除。这是因为振荡的原因不同；现在，为了达到某个角加速度，施加了过多的输入。无论姿态控制器要求什么角加速度，这种情况都会发生。

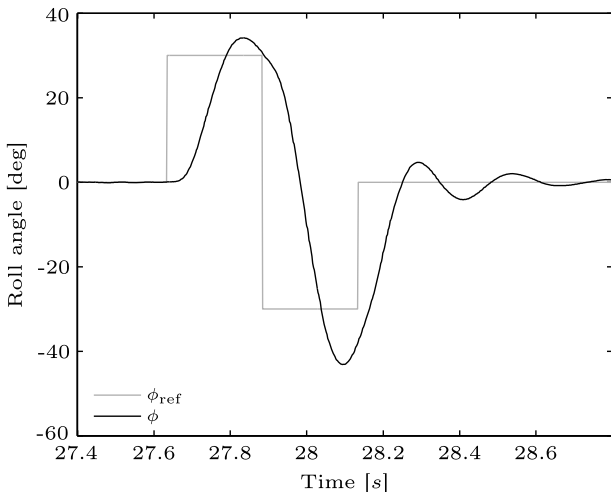


图16 无适应性的飞行，安装了保险杠，而控制效能是在未安装保险杠的情况下确定的。

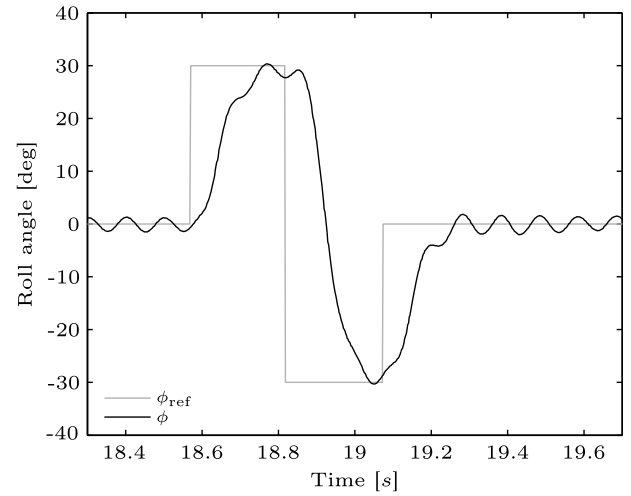
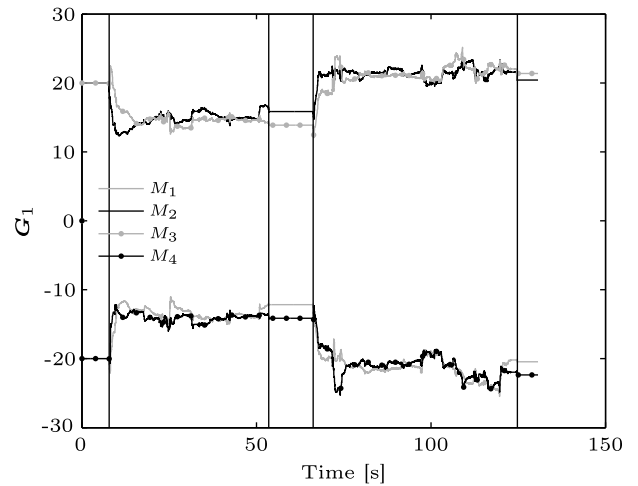
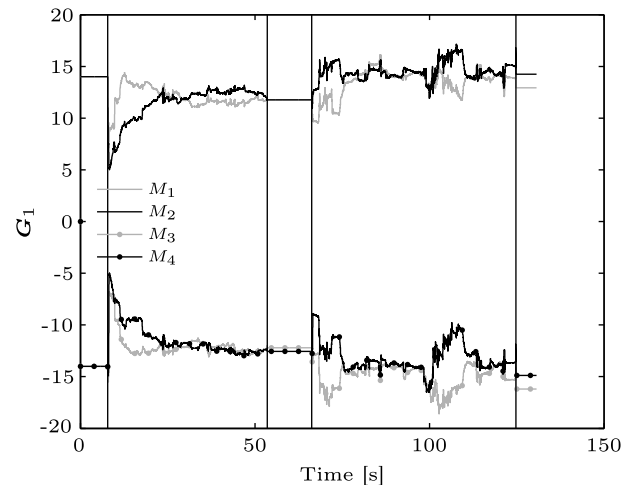


图17 无适应性飞行，未安装缓冲器，而控制效能已通过安装缓冲器确定。

我们可以得出结论，当建模的控制效能与实际控制效能不紧密对应时，性能会下降。当启用自适应算法时，图18–20展示了 G_1 矩阵的每一行如何随时间演变，这是由第五章C节所述的第二个实验产生的。同样，图21展示了 G_2 矩阵第三行的演变。每一条线代表其中一个

图18 对应滚转的 G_1 矩阵的第一行。图19 G_1 矩阵中对应音高的第二行。

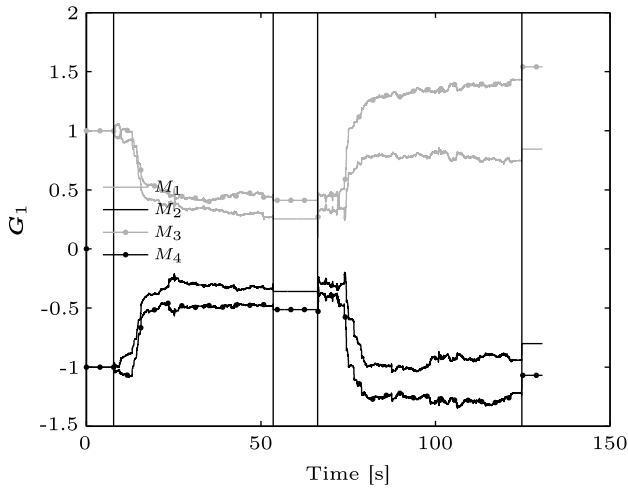


Fig. 20 Third row of the G_1 matrix corresponding to the yaw.

the elements of that row, indicating the effectiveness of that motor on the specified axis.

Note that the drone is flying in the interval of [8 54 s] and again in [66 125 s]; in between these times, the drone is landed and the bumpers are removed. This is indicated by vertical lines in the figures. A large change in effectiveness due to the addition and removal of the bumpers can be seen in the third row of the G_1 matrix, shown in Fig. 20, which corresponds to the yaw.

Also in Fig. 18, a change in effectiveness can be seen between the flights with and without bumpers. Once converged, the effectiveness values are stable with little noise. Upon takeoff and landing, the effectiveness seems to diverge for a short period of time. This is not a failure of the adaptation algorithm but merely the result of the interaction with the floor.

The controller is engaged once the pilot gives a thrust command that exceeds idle thrust. At that point, the quadrotor does not produce enough lift to take off, and so it is still standing on the floor. When the INDI controller tries to attain certain angular accelerations, the quadrotor does not rotate, and the adaptation algorithm will adapt to this. When landing, these interactions with the floor can also occur.

Notice the large difference in effectiveness between the actuators in the second part of the flight in Fig. 20. This illustrates the added value of adaptive INDI because often the actuators are assumed to perform equal to each other, whereas in this case, they do not. These differences between the actuators are also observed with the estimation method described in Sec. III.A for multiple flights. The differences may be caused by small imperfections that are not clearly visible on some of the rotors.

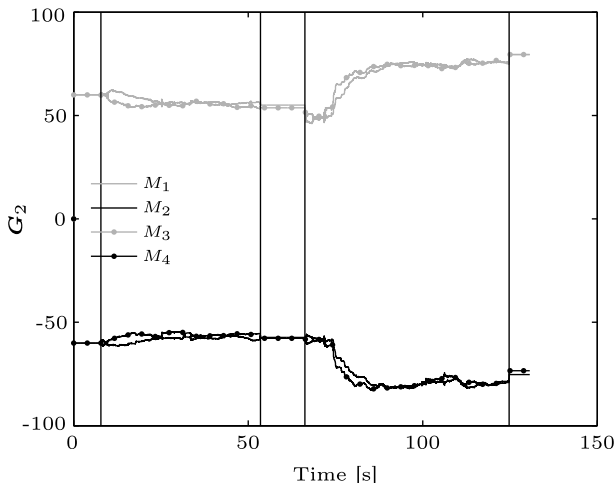


Fig. 21 Third row of the G_2 matrix corresponding to the yaw.

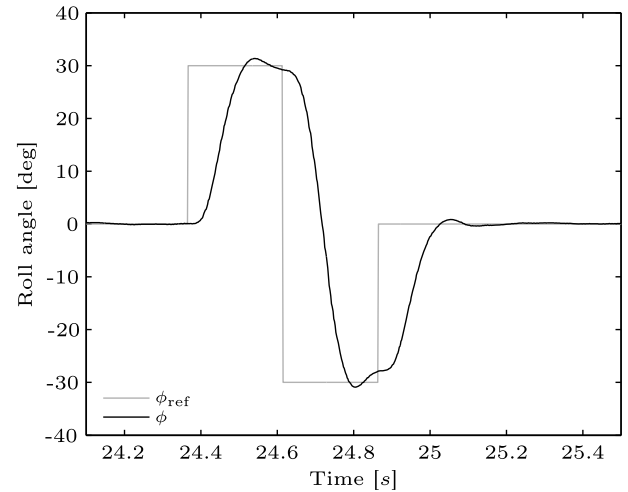


Fig. 22 Flight with adaptation, with bumpers equipped, while the control effectiveness has been determined without bumpers.

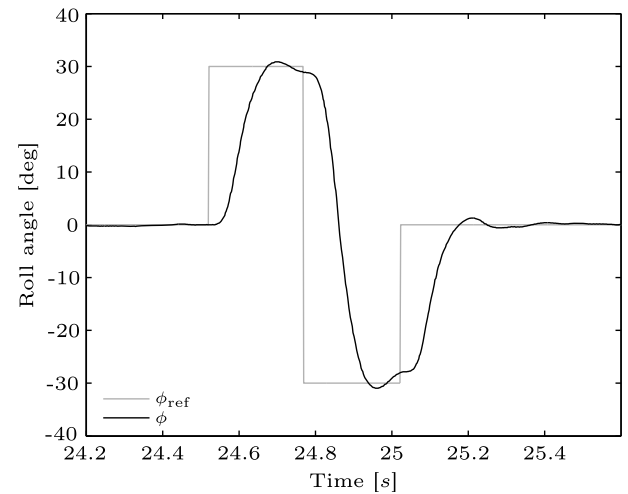


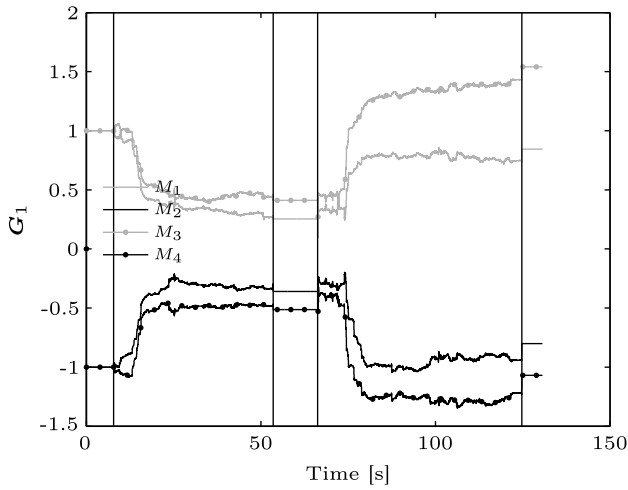
Fig. 23 Flight with adaptation, without bumpers equipped, while the control effectiveness has been determined with bumpers.

Finally, we can observe how the online parameter estimation affects the response to a roll doublet in Figs. 22 and 23. Regardless of whether the bumpers are equipped or not, or with what control effectiveness model the quadrotor starts flying, the same performance is achieved as in Sec. V.A. This shows the robustness of the adaptive algorithm against control effectiveness changes.

D. Yaw Control

Finally, consider Fig. 24. It shows for each time step the change in angular acceleration in the yaw axis, $\Delta \ddot{r}$, during the large control inputs discussed previously. A careful reader up until this point may wonder: “Is the rotor spin-up torque really significant? Can we not omit the G_2 matrix?” The figure shows the predicted change in angular acceleration based on the change in motor speeds according to Eq. (21), which is a close match. Additionally, the figure also shows the predicted change in angular acceleration if we neglect G_2 , denoted by $\Delta \ddot{r}_{\text{simple}}$. Clearly, the motor spin-up torque is very significant.

Moreover, if we try to fly with a zero G_2 matrix, the resulting oscillation is so strong that a takeoff is not possible. To fly without this matrix, we cannot use the estimated values for the control effectiveness in the yaw axis. Instead, we can take a higher effectiveness for the model parameters than in reality to avoid overshooting the reference angular acceleration due to the rotor spin-up torque that is now not taken into account. Figure 25 shows that it is possible to fly with a zero G_2 matrix, at the cost of a severe performance penalty.

图20 对应偏航的 G_1 矩阵第三行。

该行的元素，表示该电机在指定轴上的有效性。

请注意，无人机在[8 54 秒]和[66 125 秒]的时间间隔内飞行；在这些时间之间，无人机已着陆并且防撞器被移除。图中通过垂直线表示这一点。在图 20 中显示的 G_1 矩阵的第三行中，可以看到由于防撞器的加装和移除而导致的效率的巨大变化，这一行对应偏航。

在图18中，也可以看到安装缓冲器和未安装缓冲器的飞行效果存在变化。一旦收敛，效果值稳定，噪声很小。在起飞和着陆时，效果似乎在短时间内出现偏差。这不是适应算法的失败，而只是与地面交互的结果。

一旦飞行员发出超过怠速推力的推力指令，控制器就会启动。此时，四旋翼产生的升力不足以起飞，因此它仍然停在地面上。当 INDI 控制器试图达到某些角加速度时，四旋翼不会旋转，适应算法会对此进行调整。在着陆时，这些与地面的相互作用也可能发生。

注意图20中飞行后半段各执行器效能的巨大差异。这说明了自适应INDI的附加价值，因为通常假设各执行器的性能相等，而在本例中并非如此。使用第III.A节中描述的估计方法在多次飞行中也观察到了执行器之间的这些差异。这些差异可能是由某些旋翼上不易察觉的小瑕疵引起的。

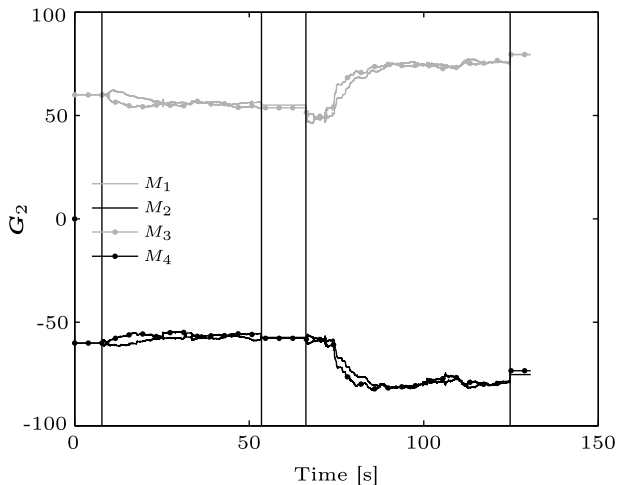
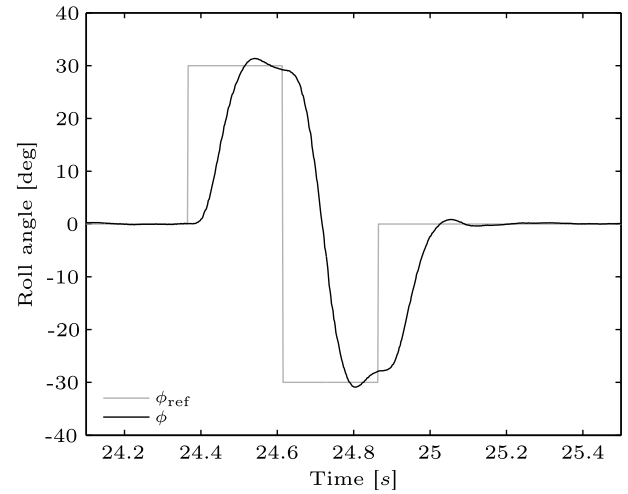
图21 对应偏航的 G_2 矩阵第三行。

图22 装有缓冲器的适应飞行，而控制效果是在未装缓冲器的情况下确定的。

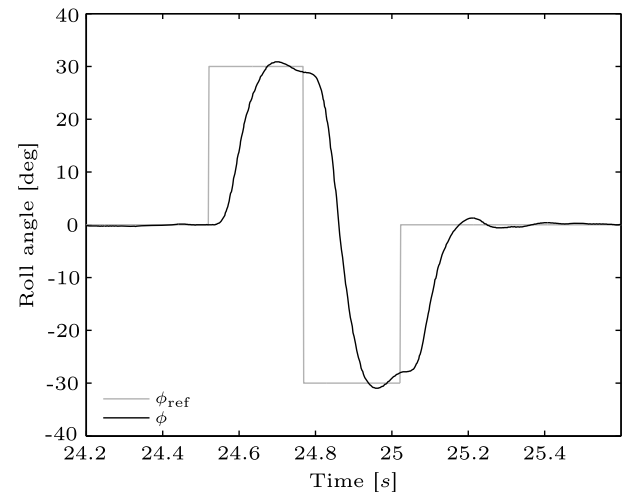


图23 装有适应性的飞行，未装备缓冲器，而控制效果已在装备缓冲器的情况下确定。

最后，我们可以在图 22 和图 23 中观察在线参数估计如何影响对滚转二重脉冲的响应。无论四旋翼是否装有缓冲器，或以何种控制有效性模型开始飞行，都能达到与第 V.A 节相同的性能。这显示了自适应算法对控制有效性变化的鲁棒性。

D. 偏航控制

最后，考虑图24。它显示了在前面讨论的大控制输入下，每个时间步中偏航轴角加速度的变化，即 Δ_r 。到目前为止，细心的读者可能会想：“旋翼加速扭矩真的重要吗？我们能否省略 G_2 矩阵？”该图显示了根据公式(21)根据电机速度变化预测的角加速度变化，结果非常接近。此外，图中还显示了如果忽略 G_2 时预测的角加速度变化，用 Δ_{r_simple} 表示。显然，电机加速扭矩非常重要。

此外，如果我们尝试使用零 G_2 矩阵飞行，产生的振荡会非常强，以至于无法起飞。若不使用该矩阵飞行，我们就不能使用偏航轴控制效能的估计值。相反，我们可以将模型参数的效能设定得高于实际值，以避免因未考虑的旋翼加速扭矩而导致参考角加速度的超调。图 25 显示，使用零 G_2 矩阵飞行是可能的，但代价是严重的性能损失。

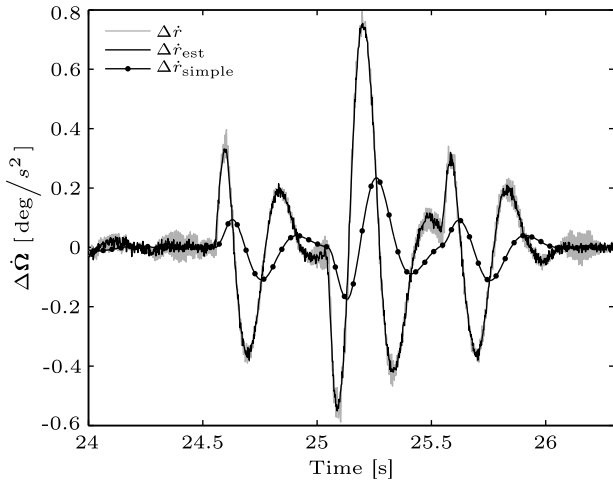


Fig. 24 Change in angular acceleration in the yaw axis along with the predicted change.

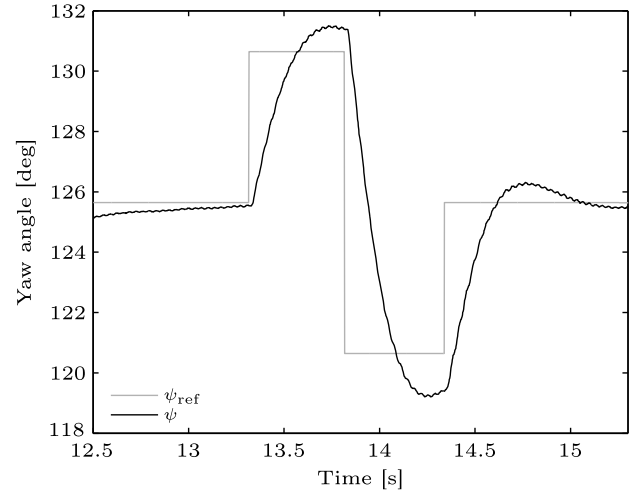


Fig. 27 Yaw angle during the doublet for the PID controller.

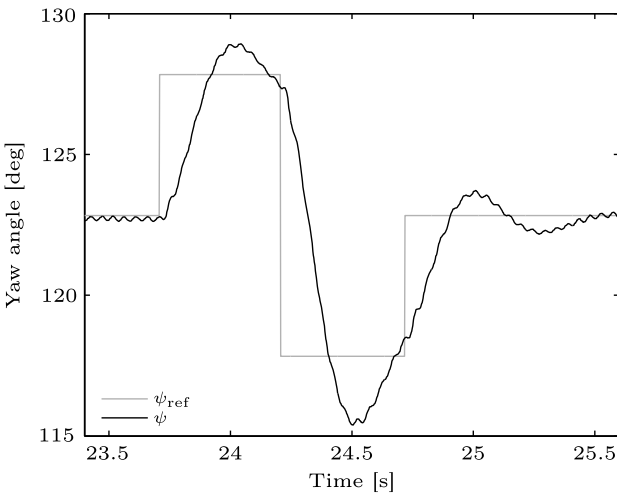


Fig. 25 Yaw angle during the doublet for the INDI controller without G_2 matrix.

If we do take the rotor angular momentum into account, Fig. 26 shows the resultant doublet response of the yaw angle. Compare this with Fig. 27, which shows the doublet response for the PID controller. The INDI controller clearly has a faster rise time and less overshoot.

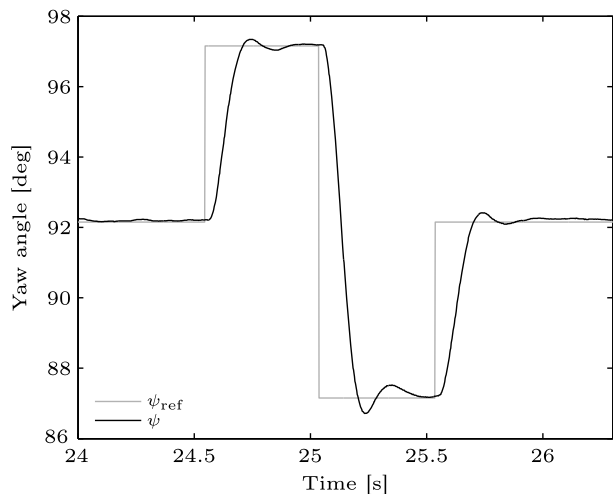


Fig. 26 Yaw angle during the doublet for the INDI controller.

VII. Conclusions

Adaptive incremental nonlinear dynamic inversion is a very promising technique for control of micro air vehicles (MAVs). Because of incorporation of the spin-up torque, fast yaw control is possible, which is typically very slow on a quadrotor. The disturbance rejection capabilities are vital when flying in windy conditions or with MAVs that have complex aerodynamics. Because unmodeled aerodynamic moments are measured with the angular acceleration, no complex aerodynamic modeling is needed. Even the control effectiveness matrices are shown to be adapted online, resulting in a controller that can handle changes in the MAV configuration and needs little effort to set up on a new platform. Only when a high-performance outer loop is required is some knowledge of the actuator dynamics needed. These properties result in a very flexible and powerful controller.

Acknowledgments

This work was financed by the Delphi Consortium. The authors would like to thank Bart Remes and the MAVLab for their support.

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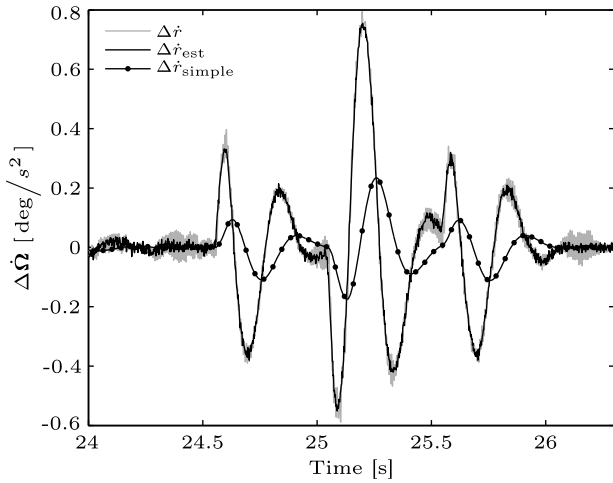


图24 偏航轴角加速度的变化及预测变化。

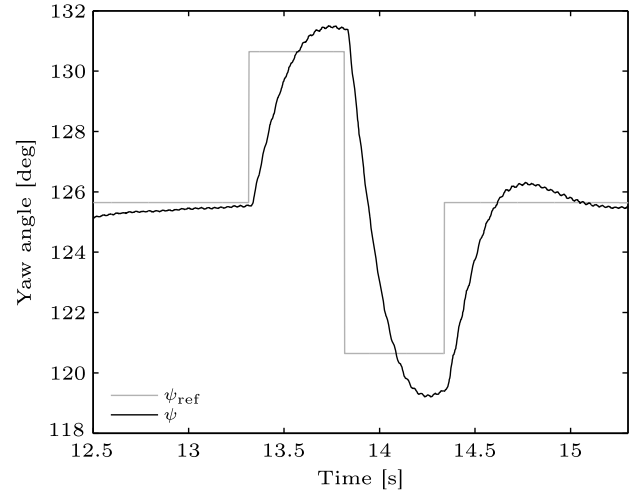
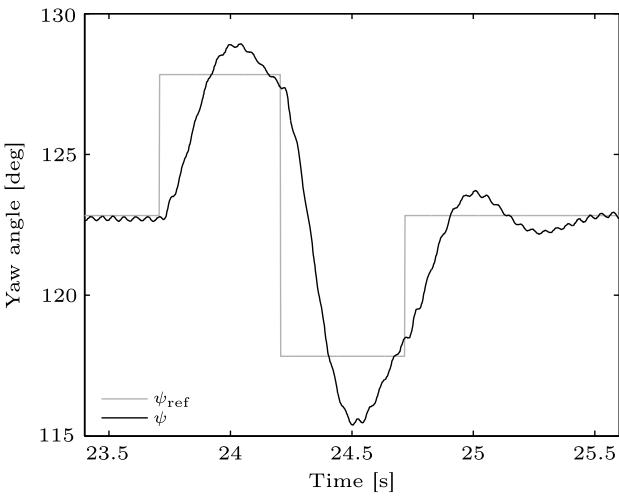


图27 PID控制器在双扰动下的偏航角。

图25 双脉冲期间无 G_2 矩阵的 INDI 控制器的偏航角。

如果我们考虑转子角动量，图26显示了偏航角的双重响应。将其与图27进行比较，图27显示了PID控制器的双重响应。INDI控制器显然具有更快的上升时间和更小的超调。

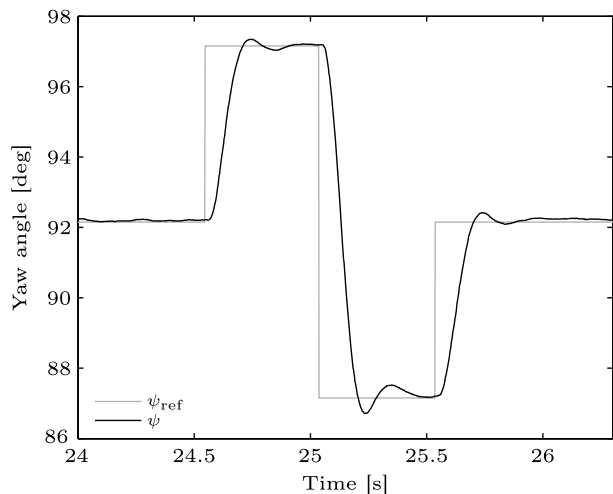


图26 双脉冲过程中INDI控制器的偏航角。

七、结论

自适应增量非线性动态反演是一种对微型飞行器 (MAV) 控制非常有前景的技术。由于引入了旋转力矩，可以实现快速的偏航控制，而在四旋翼上偏航通常很慢。当在有风的环境中飞行或操作具有复杂空气动力学的MAV时，抗扰动能力至关重要。由于未建模的空气动力矩可以通过角加速度来测量，因此无需复杂的空气动力学建模。甚至控制有效性矩阵也显示可以在线自适应，从而产生一个能够应对MAV配置变化、在新平台上设置工作量很小的控制器。仅当需要高性能外环时，才需要对执行器动态有一些了解。这些特性使得该控制器非常灵活且功能强大。

致谢

这项工作由德尔菲联盟资助。作者要感谢巴特·雷梅斯和MAV实验室的支持。

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